








Modelling Financial Markets

Chapter 1: Stylised Facts and Returns



Daniel Traian PELE

Bucharest University of Economic Studies
IDA Institute Digital Assets
Blockchain Research Center
AI4EFin Artificial Intelligence for Energy Finance
Romanian Academy, Institute for Economic Forecasting
MSCA Digital Finance

Unpredictable Returns, Predictable Volatility

What you should be able to do

- compute and aggregate simple and log returns
- measure tails with moments, quantiles and the Hill estimator
- tell zero correlation apart from independence
- choose tests whose reference distribution still holds
- estimate volatility from the open, high, low and close
- check a hypothesis proposed by an artificial intelligence (AI) assistant

Definition

- A **stylised fact** is a qualitative statistical regularity of **returns**
  - seen across assets, markets and periods Cont (2001)
- Each stylised fact remains a hypothesis, re-tested on each market and frequency Ratliff-Crain, Van Oort, Koehler & Tivnan (2025)
- Key message: returns show little linear autocorrelation
  - but their **size** is highly predictable

- Stylised facts are the benchmark every model of returns must reproduce (proofs: ▸ Appendix)
  - the i.i.d. (independent and identically distributed) Normal-distribution model fails most of them

The Stylised Facts of Cont (2001)

Group	Facts
Distribution	2. heavy tails; 3. gain/loss asymmetry; 4. aggregational Gaussianity; 7. conditional heavy tails
Dependence of returns	1. absence of linear autocorrelation
Dependence of volatility	5. intermittency; 6. volatility clustering; 8. slow decay of the autocorrelation of $\lvert r_t \rvert$; 11. asymmetry in time scales
Links to other variables	9. leverage effect; 10. volume/volatility correlation

- We re-test them on five markets, with data up to September 2026
  - we add the inference each test needs
- Early evidence: Mandelbrot (1963), Fama (1965); textbook treatment: Taylor (2008), Franke, Härdle & Hafner (2019)

Five Markets with Different Quoting Rules and Calendars

Series	Price used	Type, currency	Calendar	Sample	T
S&P 500	daily close (plus open, high, low)	price index, USD	US trading days	1990–2026	9 245
BET-TR	daily close	total return, RON	trading days of the Bucharest Stock Exchange (BVB)	Sep 2014 – 2026	2 992
Bitcoin	daily close	price, USD	every day, 24/7	Sep 2014 – 2026	4 384
EUR/RON	reference rate of the National Bank of Romania (BNR)	RON per EUR, managed float	Romanian working days	Jul 2005 – 2026	5 352
Gold	daily close	spot, USD per ounce	weekdays only	1990–2026	9 559

- Daily closes up to 18 September 2026; BET-TR: the BVB's BET index with dividends reinvested (total return)
  - daily log returns on each series' own calendar
  - T: the number of daily returns in the series

When Do Quotes Distort Measured Returns?

- **Managed float**: a market-determined exchange rate with central-bank interventions
  - the official BNR fixing replaces the EUR/RON market (interbank) quotes
  - in several years these had a lag-1 autocorrelation of -0.4 to -0.7, which came from quote errors

Simple or Log Return: What Adds Up over Time and What Adds Up across Assets

	Simple return R_t	**Log return r_t**
Definition	$(P_t + D_t)/P_{t-\mathbf{1}} - \mathbf{1}$	$\ln(1 + R_t) = \ln P_t - \ln P_{t-\mathbf{1}}$ (no dividend)
Over k periods	$\prod_{j=\mathbf{0}}^{k-\mathbf{1}}(1 + R_{t-j}) - \mathbf{1}$	$\sum_{j=\mathbf{0}}^{k-\mathbf{1}} r_{t-j}$: additive over time
Across assets	$R_{p,t} = \sum_i w_i R_{i,t}$: exact	only approximately
Range	$R_t \geq -100\%$	$(-\infty, +\infty)$
Use	portfolios, performance reporting	single-asset dynamics, central limit theorem (CLT)

- The simple return is the relative change in the value of the investment; the log return is the change in the log price
- Notation
  - P_t: the closing price on day t; D_t: the dividend paid in $(t-1, t]$
  - k: the number of periods aggregated; the index $j = 0, \ldots, k-1$ runs over them
  - $R_{i,t}$: the simple return of asset i; w_i: its weight in portfolio value, with $\sum_i w_i = 1$; $R_{p,t}$: the portfolio return
- With dividends use total-return or adjusted series (BET-TR)

Why the Log Return Is Always Below the Simple Return

- The log return is always below the simple return: $r = \ln(1 + R) < R$ for every $R \neq 0$
  - the reason: $\ln(1 + x) < x$ for every $x > -1$, $x \neq 0$, because the logarithm is concave
- The gap is negligible for small moves and large for big falls:
  - $R = +1\%$ gives $r = 0.995\%$
  - $R = -50\%$ gives $r = -69.3\%$
- Aggregation over time: the k-day log return is the sum of the k daily log returns
  - the k-day simple return needs the product of the growth factors $1 + R_{t-j}$
- Example: the price goes from 100 to 110, then to 99 (Seminar 1, A1)
  - the simple returns $+10\%$ and -10% sum to 0, yet the value falls by 1%: $1.1 \times 0.9 = 0.99$
  - the log returns 9.53% and -10.54% sum to $-1.01\% = \ln 0.99$
- Textbooks: Campbell, Lo & MacKinlay (1997), ch. 1; Franke, Härdle & Hafner (2019)

From Day to Year: the Annual Mean

- Annualisation turns the daily mean and volatility into values per year
- Notation
  - A: the number of return intervals per year; 252 for stocks and EUR/RON (trading days), 365 for crypto (traded every day), ≈ 260 for gold (the actual weekday quotes)
  - $\bar{r} = \frac{1}{T}\sum_{t=1}^{T} r_t$: the mean daily log return
  - $\hat{\sigma}$: the sample standard deviation of daily returns
- $\mu_{\text{ann}} = A\,\bar{r}$: the annual mean of **log** returns (the log growth rate); needs only a constant mean
  - the Sharpe ratio later uses the annual mean of **simple** returns, $A\bar{R}$, which is larger

From Day to Year: When the $\sqrt{A}$ Rule Holds

- $\sigma_{\text{ann}} = \sqrt{A}\,\hat{\sigma}$: the annual volatility; exact only for serially uncorrelated returns with constant variance
  - then daily variances add up, and the annual variance is $A\,\hat{\sigma}^2$
  - with autocorrelation the $\sqrt{A}$ factor is no longer correct
  - EUR/RON, lag-1 autocorrelation 0.17: annual volatility rises from 4.4% to 5.3% (Appendix; Seminar 1, A7)

CAGR and Volatility Drag: Why the Arithmetic Mean Overstates Growth

- **CAGR** (Compound Annual Growth Rate): the constant annual return that takes the price from P_0 to P_T

$$\text{CAGR} = (P_T/P_0)^{1/Y} - 1$$

  - P_0, P_T: the price at the start and at the end of the sample; Y: the number of years
  - CAGR is the annual geometric mean of the growth factors

- On the log scale, the compound growth rate is the arithmetic mean minus half the variance:

$$\ln(1 + \text{CAGR}) = \tfrac{1}{Y} \ln(P_T/P_0) \approx \mu_{\text{arith}} - \sigma^2/2$$

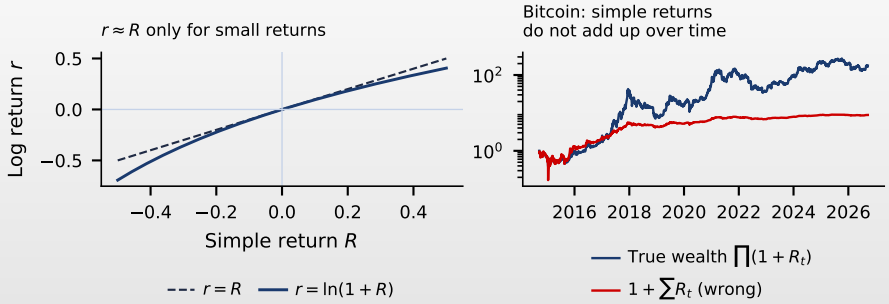
  - μ_{arith}: the annual arithmetic mean of simple returns; σ^2: their annual variance
  - $\ln(1 + \text{CAGR})$: the continuously compounded growth rate, i.e. the annual mean log return $A\bar{r}$
  - **volatility drag**: the gap between the arithmetic and the geometric mean; on the log scale it is $\approx \sigma^2/2$ and grows with the square of volatility

Volatility Drag in Numbers: S&P 500 and Bitcoin

	μ_{arith}	σ	$\sigma^2/2$	$\ln(1+\text{CAGR})$	CAGR
S&P 500 1990–2026 ($A = 252$)	9.95%	17.97%	1.61 pp	8.33%	8.69%
Bitcoin 2014–2026 ($A = 365$)	65.2%	66.2%	21.9 pp	43.1%	53.9%

- Log-scale drag, $\mu_{\text{arith}} - \ln(1+\text{CAGR})$: $9.95 - 8.33 = 1.62$ pp (percentage points) and $65.2 - 43.1 = 22.1$ pp, close to $\sigma^2/2$ (1.61 and 21.9 pp)
- Drag measured against CAGR, $\mu_{\text{arith}} - \text{CAGR}$: 1.26 pp for the S&P 500 and 11 pp for Bitcoin
  - smaller, because $\text{CAGR} = e^{\ln(1+\text{CAGR})} - 1$ lies above its log: $e^{0.431} - 1 = 53.9\%$
  - when a number is called "drag", say which of the two gaps it is
- S&P 500: $\hat{\sigma} = 1.133\%$ a day, 18.0% a year (log returns); price index, no dividends
- **Check**: gold has CAGR 6.7% and $\sigma = 16.3\%$. Estimate μ_{arith}
  - answer: $\ln 1.067 + 0.163^2/2 = 6.5\% + 1.3\% \approx 7.8\%$; with $r_f = 0$ the Sharpe ratio is $\mu_{\text{arith}}/\sigma$, so $0.48 \times 16.3\% \approx 7.8\%$ (appendix table)

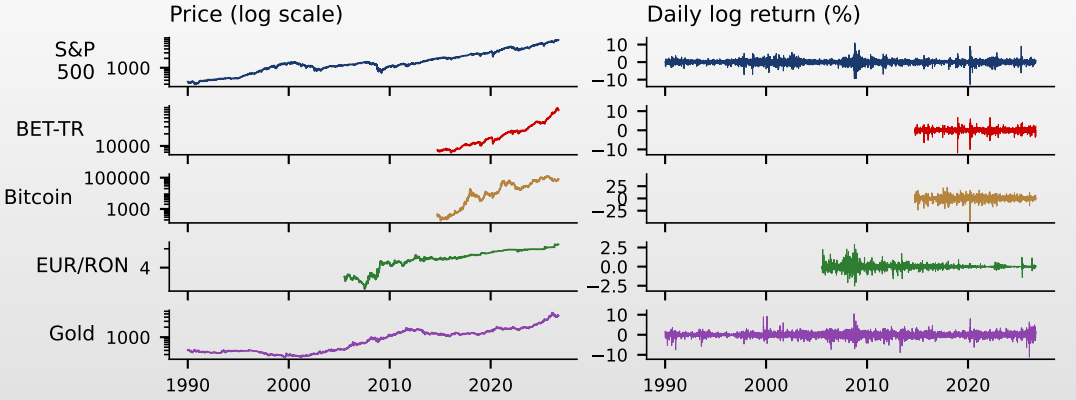
Simple and Log Returns in Practice



- Left: $r \approx R$ only for small moves
- Right: Bitcoin 2014–2026 multiplied wealth by 176.9, i.e. $\sum_t r_t = \ln 176.9 = 5.18$
  - the sum of daily log returns is $\ln(P_T/P_0)$, the log of the multiple
  - adding daily simple returns gives $1 + \sum_t R_t = 8.83$, 20 times below the true multiple

MFM_ch1_returns

Five Markets: Prices and Daily Log Returns



Prices Trend, Returns Fluctuate around a Small Mean

- **Left:** prices on log scale, trending and non-stationary
- **Right:** daily log returns fluctuate around a small mean
  - calm and turbulent periods alternate
  - extremes cluster (2008, March 2020)
- EUR/RON moves more than 1% on only 1.4% of days
- Bitcoin, 12 March 2020: log return -46.5%
  - the price is multiplied by e^{r_t}, so it falls by $1 - e^{-0.465} = 37.2\%$

MFM_ch1_returns

Fact 2, Heavy Tails: the Question and the Tools

- **The question**: are extreme days more frequent than under the Normal distribution with the same mean and variance?
- **Kurtosis** $\hat{K}$ (next slide) and the JB test: $\hat{K} > 3$ points to tails heavier than the Normal distribution's
  - both need finite moments, so they become unreliable when the tails are very heavy
- **QQ-plot**: the points $\left(\Phi^{-1}(p_i),\ z_{(i)}\right)$, with $p_i \approx (i - \frac{1}{2})/T,\ i = 1, \ldots, T$
  - $z_{(i)}$: the i-th smallest standardised return; Φ^{-1}: the quantile function of the standard Normal distribution
  - under the Normal distribution the points lie on the line $y = x$; heavy tails bend them away from it at both ends
- The **tail index** α, estimated by the Hill method from the largest values of $|r_t|$: needs no finite moments (definition)

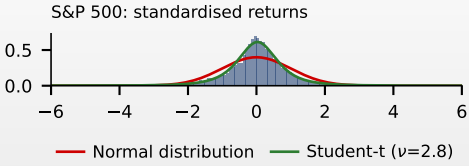
Skewness and Kurtosis: the Shape of the Distribution after Standardisation

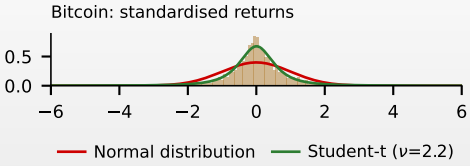
- Skewness and kurtosis describe the shape of the distribution, after removing its mean and scale

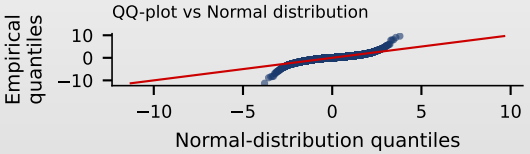
$$z_t = \frac{r_t - \bar{r}}{\hat{\sigma}}, \qquad \hat{S} = \frac{1}{T}\sum_{t=1}^{T} z_t^3, \qquad \hat{K} = \frac{1}{T}\sum_{t=1}^{T} z_t^4$$

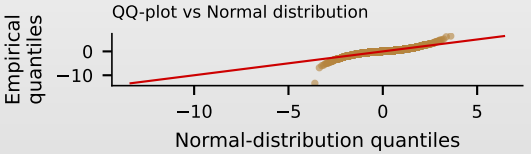
  - z_t: the standardised return, with mean 0 and standard deviation 1
  - $\hat{S}$: sample skewness; $\hat{S} < 0$: the left tail is longer, large losses dominate
  - $\hat{K}$: sample kurtosis; $\hat{K} > 3$: more mass in the centre and in the tails than under the Normal distribution
- Normal distribution: $S = 0$, $K = 3$, excess kurtosis $K - 3 = 0$

Fact 2, Heavy Tails: Histograms and Quantile–Quantile (QQ) Plots









MFM_ch1_heavy_tails

Fact 2: More Extreme Days than under the Normal Distribution

- Extreme returns are far more frequent than under the Normal distribution
  - at the same variance
- Standardised returns $(r_t - \bar{r})/\hat{\sigma}$: higher peak and fatter tails
- QQ-plots compare empirical quantiles with those of the Normal distribution
  - they bend away from the line in both tails
- A Student-t fits the body of the distribution much better than the Normal distribution
  - whether it also fits the tails is a separate question, answered with the Hill estimator below

MFM_ch1_heavy_tails

How Precisely We Estimate the Mean

- **Standard error** (SE): the standard deviation of an estimator across samples
- Mean: $\mathrm{SE}(\bar{r}) = \sigma/\sqrt{T}$, with σ the standard deviation of daily returns, estimated by $\hat{\sigma}$
  - the 95% confidence interval: $\bar{r} \pm 1.96\,\mathrm{SE}(\bar{r})$
  - daily means are 13–40 times smaller than $\hat{\sigma}$
  - so, **relative to its size**, the mean is the least precisely estimated moment
  - S&P 500: $\bar{r} = 0.033\% \pm 0.023\%$ a day (95%), $T = 9{,}245$; $\mathrm{SE}/|\bar{r}| = 0.0118/0.033 \approx 0.36$
  - higher moments have a different problem: their SE may not exist at all (section on tails)

How Precisely We Estimate Skewness and Kurtosis

- i.i.d. data from the Normal distribution: $\sqrt{T}\,\hat{S} \to N(0,6)$ and $\sqrt{T}(\hat{K}-3) \to N(0,24)$
  - $\to$: convergence in distribution as $T \to \infty$; $N(0,v)$: the Normal distribution with mean 0 and variance v
  - so $\mathrm{SE}(\hat{S}) \approx \sqrt{6/T}$ and $\mathrm{SE}(\hat{K}) \approx \sqrt{24/T}$; for the S&P 500: 0.025 and 0.051

The Delta Method

- The **delta method** gives the standard error of a function of an estimator

$$\sqrt{T}(\hat{\theta} - \theta) \to N(0, v) \implies \sqrt{T}\left(g(\hat{\theta}) - g(\theta)\right) \to N\left(0, g'(\theta)^2 v\right)$$

  - θ: a parameter, e.g. a moment; $\hat{\theta}$: its estimator; v: the asymptotic variance of $\hat{\theta}$
  - g: a differentiable function, e.g. the ratio of moments that defines $\hat{S}$; g': its derivative
  - the idea: the first-order Taylor expansion $g(\hat{\theta}) \approx g(\theta) + g'(\theta)(\hat{\theta} - \theta)$

The Delta Method: the Moments That Must Exist

- Any i.i.d. distribution: the delta method gives a Normal limit
  - for $\hat{S}$ only if $\mathbb{E}r^6 < \infty$, for $\hat{K}$ only if $\mathbb{E}r^8 < \infty$
  - $\mathbb{E}$: the expectation; $\mathbb{E}r^6 < \infty$: the sixth moment is finite
  - the variance of $\hat{S}$ involves the sixth moment, that of $\hat{K}$ the eighth
  - whether these moments exist is checked empirically in the section on tails

What the Tests Need: ARCH, Volatility Clustering, Tail Index

- **ARCH** (AutoRegressive Conditional Heteroskedasticity): today's variance depends on past returns
- **Volatility clustering**: days with large moves tend to follow days with large moves
- The **tail index** α measures how fast the probability of extreme values decays

$$P(|r_t| > x) \approx c\, x^{-\alpha} \quad \text{for large } x$$

  - x: a positive threshold; $c > 0$: a scale constant
  - a small α means a heavy tail: the probability decays slowly as x grows
  - so $\mathbb{E}|r_t|^p < \infty$ only for $p < \alpha$, where p is the order of the moment
  - the estimates in this chapter give $\alpha \approx 3$

Three Classical Tests (1/2): Normality and Autocorrelation

- **JB** (Jarque–Bera) tests normality: $JB = T\left[\hat{S}^2/6 + (\hat{K}-3)^2/24\right]$
  - the distance of skewness and kurtosis from their Normal-distribution values
- **LB** (Ljung–Box) tests autocorrelation: $Q(m) = T(T+2)\sum_{k=1}^{m} \hat{\rho}_k^2/(T-k)$
  - a weighted sum of the first m squared autocorrelations
  - $\hat{\rho}_k$: the sample autocorrelation at lag k, between r_t and r_{t-k}; here $m = 10$ lags

Three Classical Tests (2/2): ARCH Effects and How to Read the Statistics

- **ARCH-LM** tests whether squared returns are predictable from the past: TR^2 from the regression of $\hat{\varepsilon}_t^2$ on $\hat{\varepsilon}_{t-1}^2, \ldots, \hat{\varepsilon}_{t-q}^2$
  - LM: the Lagrange multiplier test; $\hat{\varepsilon}_t = r_t - \bar{r}$: the residuals; $q = 5$ lags
  - T counts observations in the auxiliary regression, after losing these lags
  - R^2: the share of the variation in $\hat{\varepsilon}_t^2$ explained by past squared residuals
- Reading: all three statistics are ≥ 0, and large values reject the null H_0
  - reference distributions: χ_2^2, χ_m^2, χ_q^2, χ_n^2: the chi-squared distribution with n degrees of freedom
  - 5% critical values: $\chi_2^2 = 5.99$, $\chi_5^2 = 11.1$, $\chi_{10}^2 = 18.3$

What Each Test Assumes and Where It Fails on Returns

Test	H_0	χ^2 limit needs	Problem here
JB, Jarque & Bera (1980)	i.i.d., Normal distribution	H_0 itself (all moments exist)	with heavy tails, near-certain rejection for large T
LB, Ljung & Box (1978)	$\rho_1 = \cdots = \rho_m = 0$	i.i.d., not only uncorrelated	volatility clustering: over-rejects
ARCH-LM, Engle (1982)	i.i.d., no ARCH effects	under the i.i.d. H_0: $\mathbb{E}r^4 < \infty$ ($\alpha > 4$)	$\alpha \approx 3 < 4$: size unreliable

- Two moment conditions for ARCH-LM, in two different situations
  - under the i.i.d. null: the χ^2_q limit needs a finite variance of r_t^2, i.e. $\mathbb{E}r^4 < \infty$
  - under GARCH-type dependence: the sample ACF of r_t^2 behind the statistic uses a standard Normal limit under the moment condition $\mathbb{E}r^8 < \infty$ ($\alpha > 8$) Davis & Mikosch (1998)
- The question is not how to run the tests
  - but whether their reference distribution holds for returns

Worked Example: JB for the S&P 500

- 5% critical value of χ^2_2: 5.99
- S&P 500, 1990–2026: $T = 9{,}245$, $\hat{S} = -0.364$, $\hat{K} - 3 = 10.88$
  - skewness term $\hat{S}^2/6 = 0.132/6 = 0.022$
  - kurtosis term $10.88^2/24 = 118.4/24 = 4.93$
  - $JB = 9{,}245 \times (0.022 + 4.93) \approx 45{,}800$, against the critical value 5.99
- Almost all of the statistic comes from the kurtosis term
  - the problem is the tails, not the asymmetry

What a JB Rejection Does and Does Not Show

- Under the null all moments exist
  - so the p-value is valid for the **joint** null "i.i.d. and Normal distribution"
  - heavy tails make JB grow with T
  - this is power, not a broken p-value
- A rejection does not say which part fails
  - it may be the marginal distribution or independence
  - tests of the marginal distribution with HAC (heteroskedasticity and autocorrelation consistent) variances:

▸ Appendix

Robust Alternative: Quantile-Based Skewness and Kurtosis

- Two quantile-based measures Kim & White (2004), which need no moments
  - q_p: the quantile of order p of returns, the value below which a share p of days lies; $q_{0.5}$: the median
- **Hinkley skewness** compares the lengths of the two tails around the median

$$\frac{q_{0.95} + q_{0.05} - 2q_{0.5}}{q_{0.95} - q_{0.05}}$$

  - the numerator is $(q_{0.95} - q_{0.5}) - (q_{0.5} - q_{0.05})$: right tail minus left tail
  - it lies in $[-1, 1]$; a negative value indicates a longer left tail
- **Crow–Siddiqui excess kurtosis**, denoted K_{CS}, compares the width of the tails with the width of the centre

$$K_{CS} = \frac{q_{0.975} - q_{0.025}}{q_{0.75} - q_{0.25}} - 2.91$$

  - 2.91: the value of the ratio for the Normal distribution; a positive value indicates heavier tails
  - both measures are 0 for the Normal distribution

Confidence Intervals from a Moving-Block Bootstrap

- 95% confidence intervals (CI) from a **moving-block bootstrap**
  - it resamples blocks of 20 consecutive days (1,000 resamples)
  - the measure is recomputed on each resampled series
  - the blocks keep the dependence within them
- Valid for stationary, weakly dependent returns with a positive density at these quantiles
- A CI that excludes 0 indicates a significant departure from the Normal distribution

Heavy Tails Are Confirmed without Moments

	Moment skew.	Hinkley [95% CI]	Excess kurt.	Crow–Siddiqui excess [95% CI]
S&P 500	−0.36	−0.064 [−0.092, −0.041]	10.9	1.54 [1.29, 1.79]
BET-TR	−1.40	−0.036 [−0.085, 0.025]	17.8	1.06 [0.73, 1.43]
Bitcoin	−0.70	−0.018 [−0.069, 0.032]	12.0	2.25 [1.88, 2.64]
EUR/RON	0.73	0.041 [−0.022, 0.093]	15.1	4.87 [3.74, 6.13]
Gold	−0.23	0.004 [−0.025, 0.030]	9.3	1.47 [1.29, 1.73]

- Quantile (Crow–Siddiqui) excess kurtosis is positive in all five markets
  - no CI contains 0
- Quantile-based negative skewness is established only for the S&P 500
  - BET-TR's moment skewness −1.40 comes from a few days (19 December 2018)

MFM_ch1_inference

Summary Statistics: Five Markets

	S&P 500	BET-TR	Bitcoin	EUR/RON	Gold
Observations	9245	2992	4384	5352	9559
Mean (% per day)	0.033	0.077	0.118	0.007	0.025
Std. dev. (% per day)	1.13	0.98	3.49	0.28	1.01
Annualised volatility (%)	18.0	15.5	66.6	4.4	16.3
Skewness	−0.36	−1.40	−0.70	0.73	−0.23
Excess kurtosis	10.9	17.8	12.0	15.1	9.3
Min / max log return (%)	−12.8 / 11.0	−11.9 / 6.7	−46.5 / 22.5	−2.5 / 2.9	−11.1 / 10.4
Jarque–Bera	45805	40433	26482	51575	34732
Ljung–Box $Q(10)$, r_t	91.6	70.0	20.2	221.7	32.1
Ljung–Box $Q(10)$, r_t^2	8018	714	213	2420	1049
ARCH-LM (5 lags)	2229	371	115	871	483

- Descriptive: heavy tails everywhere (excess kurtosis 9.3–17.8)
  - $Q(10)$ on r_t^2 is 10–90 times larger than on r_t
  - the size of returns is far more predictable than their direction
- Inferential: all i.i.d. p-values are below 0.03
  - but they are valid only for i.i.d. returns with the required moments ($\alpha > 4$ for ARCH-LM)

MFM_ch1_summary_statistics

The Student-t Distribution: a Heavy-Tailed Model

- The Student-t distribution, Praetz (1972): a heavy-tailed model for returns, with density

$$f(x) \propto \left(1 + \frac{(x-\mu)^2}{\nu s^2}\right)^{-(\nu+1)/2}$$

- Notation
  - $f(x)$: the density at x; $\propto$: "proportional to", i.e. without the normalising constant
  - μ: the centre of the distribution; $s > 0$: the scale parameter
  - $\nu > 0$: the degrees of freedom; a small ν means heavy tails; as $\nu \to \infty$ we get the Normal distribution

The Student-t Distribution: Tails and Moments

- For a variable X with this distribution
  - tail $P(X > x) \propto x^{-\nu}$: moments exist only of order $< \nu$
  - variance $s^2\nu/(\nu - 2)$ for $\nu > 2$
  - excess kurtosis $6/(\nu - 4)$ for $\nu > 4$

Extreme Days: Observed, under the Normal Distribution and under the Student-t

	S&P 500	BET-TR	Bitcoin	EUR/RON	Gold
Days beyond $3\hat{\sigma}$ (%)	1.55	1.44	1.82	2.06	1.42
Days beyond $5\hat{\sigma}$ (count)	30	12	10	29	26
Expected under the Normal distribution, $> 5\hat{\sigma}$	0.005	0.002	0.003	0.003	0.005
Student-t degrees of freedom $\hat{\nu}$ (maximum likelihood)	2.8	3.1	2.2	1.2	2.9

- Under the Normal distribution
  - 0.27% of days lie beyond 3σ (observed 5–8 times more)
  - one 5σ day occurs every 6,900 years of trading (observed 10–30)
- $\hat{\nu}$ is fitted by **maximum likelihood** to the whole distribution: it picks the μ, s and ν that make the observed data most probable
  - the many small returns dominate the fit, so a poor fit in the centre moves $\hat{\nu}$
  - EUR/RON: $\hat{\nu} = 1.2$ reflects that 57% of daily changes are below 0.1%, plus rare jumps

The Hill Estimator: the Tail Index from the Largest Observations

- The tail index α of $X = |r_t|$
  - $\alpha > 2$: finite variance
  - $\alpha > 4$: finite kurtosis
  - power laws were first argued by Mandelbrot (1963) and Fama (1965)
- The **Hill estimator**, Hill (1975): estimates α from the k largest observations only
  - order them from largest to smallest: $X_{(1)} \geq X_{(2)} \geq \cdots \geq X_{(n)}$
  - $X_{(i)}$: the i-th largest value; n: the number of observations
  - k: the number of tail observations; $X_{(k+1)}$ acts as the threshold where the tail starts
- The **Pareto tail**, $P(X > x) = (x/u)^{-\alpha}$ for $x \geq u$, is the exact power law; u is the threshold

The Hill Estimator: the Inverse Mean Log Exceedance

- The Hill estimator is the inverse of the mean log exceedance over the threshold

$$\hat{\alpha}_k = \left[\frac{1}{k} \sum_{i=1}^{k} \ln X_{(i)} - \ln X_{(k+1)} \right]^{-1} = \left[\frac{1}{k} \sum_{i=1}^{k} \ln \frac{X_{(i)}}{X_{(k+1)}} \right]^{-1}$$

  - $\ln(X_{(i)}/X_{(k+1)})$: how far the i-th value exceeds the threshold, on a log scale
  - for a Pareto tail these exceedances average $1/\alpha$
  - large exceedances mean a heavy tail, hence a small $\hat{\alpha}_k$

- Standard error $\mathrm{SE}(\hat{\alpha}_k) \approx \hat{\alpha}_k/\sqrt{k}$; 95% CI: $\hat{\alpha}_k \pm 1.96\,\mathrm{SE}$
  - it assumes independent tail observations and a Pareto tail
  - clustering of extremes makes it larger (Appendix)

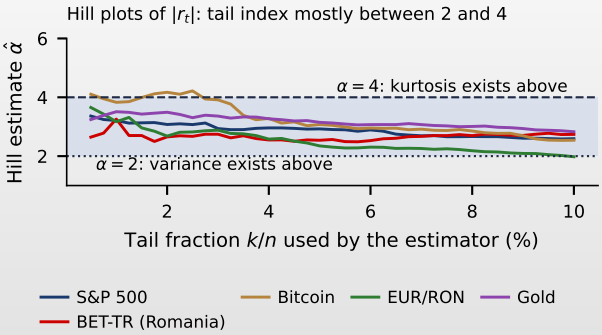
The Hill Estimator by Hand: a Five-Number Example

- Toy data: the five largest $|r_t|$ of a sample, in %: 4.0, 3.0, 2.5, 2.2, 2.0
  - take $k = 4$, so the threshold is $X_{(5)} = 2.0$
- Log exceedances $\ln(X_{(i)}/X_{(5)})$, $i = 1, \ldots, 4$
  - $\ln 2 = 0.693$, $\ln 1.5 = 0.405$, $\ln 1.25 = 0.223$, $\ln 1.1 = 0.095$
  - mean $1.417/4 = 0.354$, so $\hat{\alpha}_4 = 1/0.354 = 2.82$
- SE $\approx 2.82/\sqrt{4} = 1.41$; 95% CI $[0.06, 5.59]$
  - with 4 tail points the estimate says almost nothing; this is why the tables use k in the hundreds
- **Check**: double every value (a market twice as volatile). What happens to $\hat{\alpha}_4$?
  - answer: nothing; the ratios $X_{(i)}/X_{(5)}$ are unchanged, so a change of scale does not move the tail index
- Full worked exercise on market data: Seminar 1, A5

Choosing k: between Variance and Bias

- **Choice of** k, i.e. of the threshold $u = X_{(k+1)}$
  - small k: few tail observations, so a large variance (SE $\approx \alpha/\sqrt{k}$)
  - large k: observations from the centre enter, where the tail is no longer Pareto, so $\hat{\alpha}$ is biased
  - read the level where the **Hill plot** ($\hat{\alpha}_k$ against k) is flat
- The chapter's rule, fixed before estimation: $k = 2.5\%$ of n, the middle of the 1–5% range where the Hill plots are roughly flat
- Robustness: we also report the smallest and the largest $\hat{\alpha}_k$ for k between 1% and 5% of n
- The Student-t has tail index $\alpha = \nu$, but fitted to the whole distribution it misdescribes the tails: for EUR/RON $\hat{\alpha} = 2.8$ (Hill) while $\hat{\nu} = 1.2$

Hill Plots for Five Markets



- $\hat{\alpha}$ for $|r_t|$ at $k = 2.5\%$ of n
  - S&P 500 3.1, BET-TR 2.7, Bitcoin 4.2, EUR/RON 2.8, gold 3.3
- For $k = 1$–5%: between 2.3 and 3.5 except Bitcoin (3.1–4.2)
- With the same threshold u for both tails, the loss tail is heavier for BET-TR (2.2 versus 3.1) and Bitcoin (2.7 versus 3.4)
- For EUR/RON, the gain tail is heavier
- Tails are modelled with extreme value theory (Chapter 7)

MFM_ch1_heavy_tails

Tail Indices with Confidence Intervals

$\lvert r_t \rvert$, $k = 2.5\%$ of n	n	k	$\hat{\alpha}$	SE i.i.d.	95% CI (i.i.d.)	SE block bootstrap
S&P 500	9245	231	3.07	0.20	[2.68, 3.47]	0.33
BET-TR	2992	74	2.67	0.31	[2.06, 3.28]	0.45
Bitcoin	4384	109	4.22	0.40	[3.43, 5.01]	0.47
EUR/RON	5352	133	2.82	0.24	[2.34, 3.30]	0.28
Gold	9559	238	3.31	0.21	[2.89, 3.73]	0.24

- Moving-block bootstrap (blocks of 20 days, 500 resamples) keeps the clustering of extremes
  - SEs are 1.1–1.6 times the i.i.d. ones

MFM_ch1_inference

The Fourth Moment Very Likely Does Not Exist

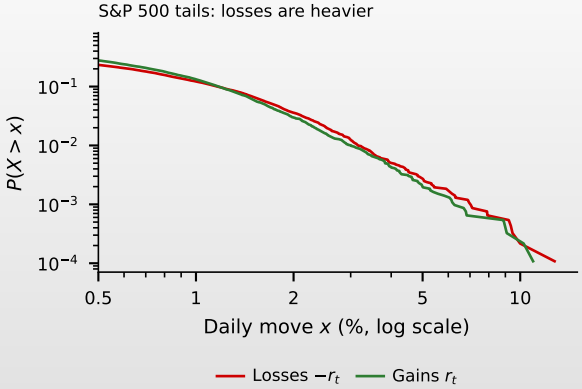
- $\alpha = 4$ is outside every interval except Bitcoin's
  - also when the interval uses the bootstrap SE
  - the fourth moment very likely does not exist
- $\alpha = 2$ is outside every i.i.d. interval
  - with the bootstrap SE only BET-TR reaches it: the lower limit $\hat{\alpha} - 1.96\,\mathrm{SE} = 2.67 - 1.96 \times 0.45 = 1.79$
- Bitcoin's 4.2 uses a threshold where its Hill plot is not yet flat
  - at a common threshold its two tails give 2.71 and 3.36 (Appendix)

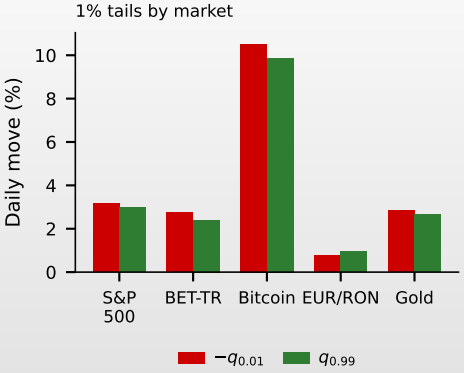
MFM_ch1_inference

Fact 3: How We Measure and Test the Gain/Loss Asymmetry

- **The question**: are large losses more frequent than gains of the same size?
- The empirical **survival function** $\hat{P}(X > x) = \frac{1}{T} \#\{t : X_t > x\}$, with $X_t = -r_t$ for losses and $X_t = r_t$ for gains
  - on a log–log scale a Pareto tail $P(X > x) = (x/u)^{-\alpha}$ is a straight line with slope $-\alpha$
- **The statistic** $\Delta = -q_{0.01} - q_{0.99}$: $\Delta > 0$ means the 1% loss is larger than the 1% gain
  - $-q_{0.01}$ is the 1% **VaR** (Value-at-Risk), the daily loss threshold exceeded with probability 1% (Chapter 7)
- **The test**: $H_0 : \Delta = 0$ is rejected if the 95% moving-block bootstrap CI (blocks of 20 days) excludes 0; the blocks keep the clustering of extreme days
- For the far tails: the Hill tail indices α_- (losses) and α_+ (gains), at the same threshold u

Fact 3: Gain/Loss Asymmetry in the Data





MFM_ch1_heavy_tails

Fact 3: Extreme Losses Exceed Gains, but the Gap Is Rarely Significant

- Left: the S&P 500 survival functions on a log–log scale; the loss curve lies above the gain curve
- Right: $-q_{0.01}$ (1% VaR, red) and $q_{0.99}$ (green) in the five markets
- $\Delta > 0$ in 4 of 5 markets, but the 95% CI contains 0 in all four
  - S&P 500: $\Delta = 3.15 - 2.97 = 0.18$ pp, CI $[-0.12,\ 0.49]$ pp
- EUR/RON: $\Delta = -0.17$ pp, CI $[-0.35,\ -0.08]$ pp: leu depreciations (up-moves) are significantly larger
- At a common threshold the loss tail is heavier for BET-TR ($\hat{\alpha}_- = 2.20$ versus $\hat{\alpha}_+ = 3.06$, $z = -2.7$ with block-bootstrap SEs) and, marginally, for Bitcoin (2.71 versus 3.36, $z = -1.9$)
- A Normal-distribution model with the right variance underestimates both extreme quantiles

MFM_ch1_heavy_tails

Facts 1 and 6: the Questions, the Statistics and the Bands

- **Fact 1**: does the direction of returns repeat? Statistic: $\hat{\rho}_k(r)$, the autocorrelation between r_t and r_{t-k} at lag k (days)
- **Fact 6**: does their size repeat (volatility clustering)? Statistics: $\hat{\rho}_k(|r|)$ and $\hat{\rho}_k(r^2)$
  - r_t^2 links directly to the variance, as in ARCH and GARCH models
  - $|r_t|$ needs fewer moments: the ACF of $|r_t|$ has a Normal limit for $\alpha > 4$, that of r_t^2 only for $\alpha > 8$ Davis & Mikosch (1998)
- **The test** at each lag: $H_0 : \rho_k = 0$, rejected when $\hat{\rho}_k$ leaves a 95% band
  - the band is centred at 0, the value under H_0: it is a rejection band, not a confidence interval (which would be centred at $\hat{\rho}_k$)
- Two bands: i.i.d., $\pm 1.96/\sqrt{T}$, and robust to volatility clustering, $\pm 1.96\sqrt{\hat{\tau}_k}$ (next slide)

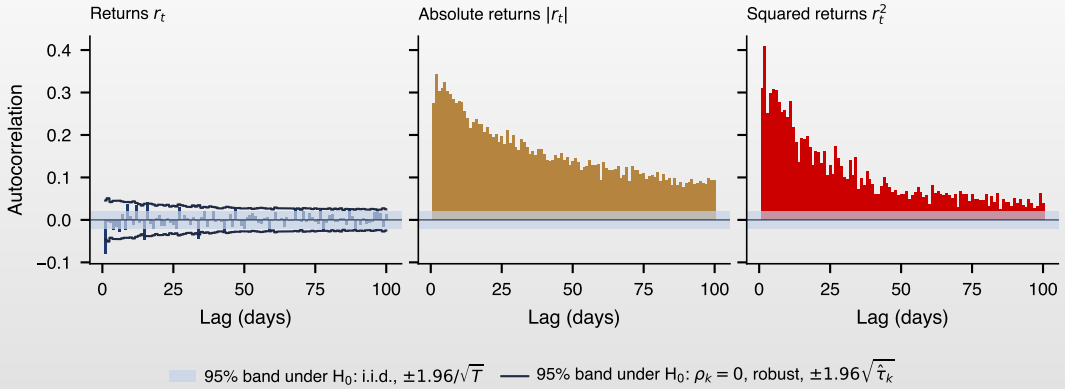
The ACF under Conditional Heteroskedasticity

- **ACF** (AutoCorrelation Function): the sample autocorrelations $\hat{\rho}_k$, plotted against the lag k

$$\hat{\rho}_k = \frac{\sum_t x_t x_{t-k}}{\sum_t x_t^2}, \qquad x_t = r_t - \bar{r}$$

  - x_t: the demeaned return; $\hat{\rho}_k \in [-1, 1]$
  - $\hat{\rho}_k > 0$: a day above the mean tends to be followed, k days later, by a day above the mean
- Uncorrelated returns whose conditional variance moves: $\hat{\rho}_k \approx N(0, \tau_k)$ Romano & Thombs (1996)
  - τ_k: the variance of $\hat{\rho}_k$, estimated by $\hat{\tau}_k = \sum_t x_t^2 x_{t-k}^2 / (\sum_t x_t^2)^2$
  - i.i.d.: $T\tau_k = 1$, which gives the classical band $\pm 1.96/\sqrt{T}$
  - volatility clustering: $T\tau_k > 1$; the classical band is too narrow and flags autocorrelations as significant too often

Facts 1 and 6: Uncorrelated but Not Independent



MFM_ch1_volatility_clustering

Facts 1 and 6: Direction Does Not Repeat, Size Does

- Left, r_t (S&P 500, 1990–2026): $\hat{\rho}_1 = -0.08$
  - 20 of 100 lags leave the i.i.d. band (blue area), but only 7 leave the robust band (lines)
  - at 5%, about 5 lags in 100 leave the band by chance: apart from lag 1, Fact 1 holds
- Middle and right: $\hat{\rho}_k(|r|)$ and $\hat{\rho}_k(r^2)$ are positive at all 100 lags ($\hat{\rho}_1(|r|) = 0.27$): Fact 6
- Returns are uncorrelated but not independent: independence requires zero correlation of **every** function of returns
- The model $r_t = \sigma_t \varepsilon_t$ (GARCH, Chapter 5)
  - σ_t: the conditional volatility on day t, driven by past returns; ε_t: i.i.d. shocks with mean 0 and variance 1

MFM_ch1_volatility_clustering

The Robust Portmanteau Test $\tilde{Q}(m)$

- Robust t-statistic: $\hat{\rho}_k / \sqrt{\hat{\tau}_k}$, compared with ± 1.96
- Robust portmanteau test Lobato, Nankervis & Savin (2001): null hypothesis $\rho_1 = \cdots = \rho_m = 0$

$$\tilde{Q}(m) = \sum_{k=1}^{m} \frac{\hat{\rho}_k^2}{\hat{\tau}_k}$$

  - the sum of the squared robust t-statistics for the first m lags, so $\tilde{Q}(m) \geq 0$
  - reference distribution χ^2_m; large values reject the null
  - its assumptions: Appendix
- The difference from the Ljung–Box $Q(m)$
  - $Q(m)$ divides each $\hat{\rho}_k^2$, approximately, by the i.i.d. variance $1/T$
  - $\tilde{Q}(m)$ divides by the estimated variance $\hat{\tau}_k$

With Robust Standard Errors, Only the S&P 500 and EUR/RON Stay Significant

Lag 1	S&P 500	BET-TR	Bitcoin	EUR/RON	Gold
$\hat{\rho}_1$	−0.079	0.061	−0.022	0.170	−0.011
$T\hat{\tau}_1$	5.00	4.35	2.89	6.85	2.43
t-statistic, i.i.d. / robust	−7.6 / −3.4	3.3 / 1.6	−1.5 / −0.9	12.4 / 4.7	−1.1 / −0.7

- S&P 500: $Q(10) = 91.6$ but $\tilde{Q}(10) = 19.0$ (p $= 0.041$)
- With robust SEs only the S&P 500 ($t = -3.4$) and EUR/RON ($t = 4.7$) keep a significant lag-1 autocorrelation
  - it is economically small

MFM_ch1_inference

Heavy Tails from Random Volatility: Mixtures of Normal Distributions

- The model: the return is a Normal shock times a random volatility

$$r_t = \sigma_t \varepsilon_t, \qquad \varepsilon_t \sim N(0,1) \text{ independent of } \sigma_t$$

- The kurtosis of returns is at least 3

$$K = \frac{\mathbb{E}[r_t^4]}{\mathbb{E}[r_t^2]^2} = 3\,\frac{\mathbb{E}[\sigma_t^4]}{\mathbb{E}[\sigma_t^2]^2} \;\geq\; 3$$

  - the middle equality: $\mathbb{E}[\varepsilon_t^4] = 3$ and $\mathbb{E}[\varepsilon_t^2] = 1$, and σ_t and ε_t are independent
  - the inequality is Jensen's inequality ($\mathbb{E}[X^2] \geq \mathbb{E}[X]^2$) with $X = \sigma_t^2$
  - equality only if σ_t is constant; any variation in volatility creates excess kurtosis

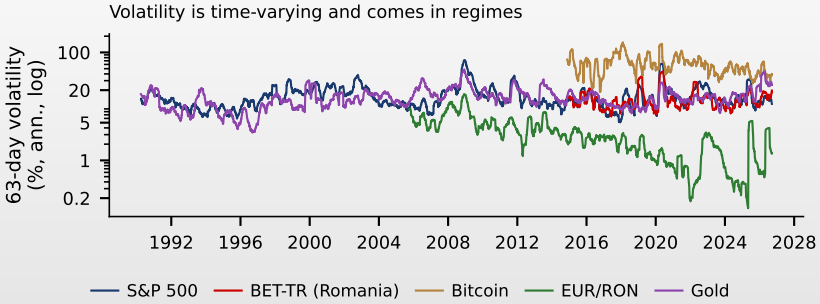
A Numerical Example: Kurtosis 8.33 from Normal Shocks

- Example: $\sigma = 1\%$ on 90% of days and 3% on 10% of days

$$K = 3 \cdot \frac{0.9 \cdot 1 + 0.1 \cdot 81}{(0.9 + 0.1 \cdot 9)^2} = 3 \cdot \frac{9.0}{3.24} = 8.33$$

  - numerator: $\mathbb{E}[\sigma^4] = 0.9 \cdot 1^4 + 0.1 \cdot 3^4$, in units of $(1\%)^4$
  - denominator: $\mathbb{E}[\sigma^2]^2$, with $\mathbb{E}[\sigma^2] = 0.9 \cdot 1^2 + 0.1 \cdot 3^2 = 1.8$
  - the excess kurtosis is 5.33, although each day's return, given σ, is Normally distributed
- This links heavy tails (Fact 2) to volatility clustering (Fact 6)
  - a mixture drawn independently every day has heavy tails but no clustering (Seminar 1, A4 and A9)

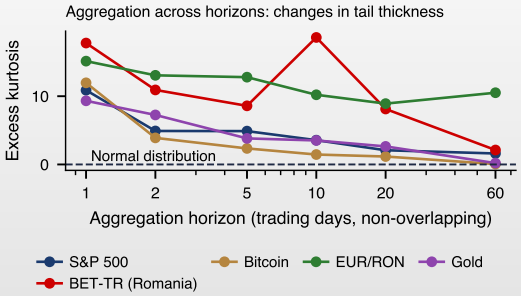
Fact 5: Intermittency in the Data



- Bursts of high volatility separated by calm periods: constant-volatility models cannot produce them
- 63-day rolling volatility $\sqrt{A}\times$ the standard deviation over the window, log scale
  - common spikes: 2008–2009, March 2020
- EUR/RON volatility collapsed after June 2013, under the BNR's managed float

MFM_ch1_volatility_clustering

Fact 4: Aggregational Gaussianity



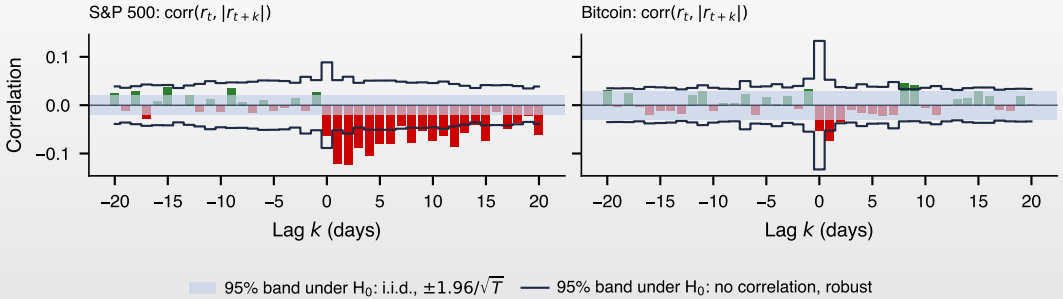
- **The question**: do h-day returns, $r_t(h) = r_t + \cdots + r_{t-h+1}$, approach the Normal distribution as h grows?
- Statistic: the excess kurtosis $\hat{K}(h) - 3$ of non-overlapping block sums; n_h: their number
- Test: under H_0 ($r_t(h)$ i.i.d. Normal) all moments exist and $z = (\hat{K}(h) - 3)/\sqrt{24/n_h} \approx N(0,1)$
- The chart: $\hat{K}(h) - 3$ against h in trading days; dotted line: the Normal distribution
- At $h = 60$ (a quarter for the S&P 500, two months for Bitcoin): S&P 500 $10.9 \to 1.6$ ($n_h = 154$, $z = 4.1$), Bitcoin $12.0 \to 0.1$ ($n_h = 73$, $z = 0.1$)
- Normality is still rejected at $h = 60$ for the S&P 500, BET-TR ($z = 3.0$) and EUR/RON (10.5, $z = 20$), but not for Bitcoin and gold
- The CLT works slowly here: daily returns are dependent and heavy-tailed

MFM_ch1_heavy_tails

Fact 9: the Leverage Effect, Statistic and Test

- **The question**: does volatility rise more after falls than after rises?
- Statistic: $L(k) = \text{corr}(r_t, |r_{t+k}|)$, the correlation of today's return with the size of the return k days later
  - $L(k) < 0$ for $k > 0$: after falls $|r_t|$ grows, i.e. volatility rises
  - $k < 0$ is a control: the size of past returns should not predict the sign of today's
- **The test**: $H_0 : L(k) = 0$, with $z = \hat{L}(k)/\text{SE}$ compared with ± 1.96
  - i.i.d. SE $= 1/\sqrt{T}$: ignores volatility clustering, so the $\pm 1.96/\sqrt{T}$ band is too narrow
  - robust SE $= \sqrt{\hat{\tau}}$, as for the ACF: $\hat{\tau} = \sum_t x_t^2 y_{t+k}^2 / \left(\sum_t x_t^2 \sum_t y_t^2 \right)$, with $x_t = r_t - \bar{r}$ and $y_t = |r_t| - \overline{|r|}$

Fact 9: The Leverage Effect



MFM_ch1_leverage_volume

Fact 9: Volatility Rises after Falls

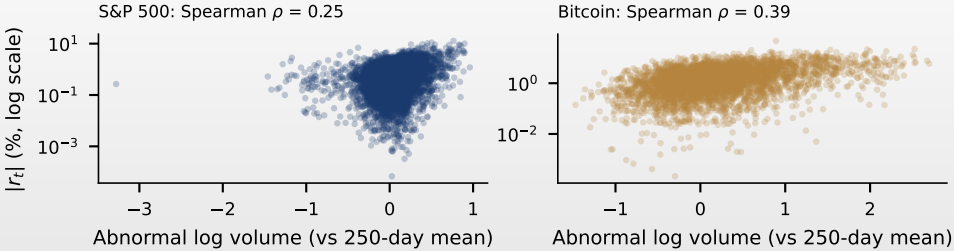
- The chart: $\hat{L}(k)$ for k from -20 to 20; blue area: the i.i.d. band; lines: the robust band
- S&P 500: $L(1) = -0.12$ ($z = -4.6$, robust SE), still -0.07 at $k = 10$; for $k < 0$ no value is significant
- BET-TR: $L(1) = -0.12$ ($z = -2.8$); Bitcoin: -0.07 ($z = -2.7$); gold: -0.02 ($z = -0.9$, not significant)
- EUR/RON: reversed, $L(1) = +0.10$ ($z = 2.7$): volatility rises after leu depreciations
- Possible explanations: the financial leverage of firms Christie (1982); the retarded volatility model Bouchaud, Matacz & Potters (2001)
- Asymmetric GARCH models: GJR-GARCH (Glosten–Jagannathan–Runkle) and EGARCH (exponential GARCH) (Chapter 5)

MFM_ch1_leverage_volume

Fact 10: Volume and Volatility, Statistic and Test

- **The question**: are days of unusually high volume also days of large price moves?
- Abnormal volume: $\tilde{v}_t = \ln V_t - \frac{1}{250}\sum_{j=0}^{249} \ln V_{t-j}$
  - V_t: the volume traded on day t; the moving average removes the long-run trend in volume
- Statistic: the Spearman rank correlation $\rho_S = \mathrm{corr}\big(\mathrm{rank}(\tilde{v}_t),\ \mathrm{rank}(|r_t|)\big) \in [-1, 1]$
  - ranks make it robust to the heavy tails of both series and to monotone non-linear relations
- **The test**: $H_0 : \rho_S = 0$ is rejected if the 95% moving-block bootstrap CI (blocks of 20 days) excludes 0; the blocks keep the persistence of both series

Fact 10: Volume and Volatility Move Together



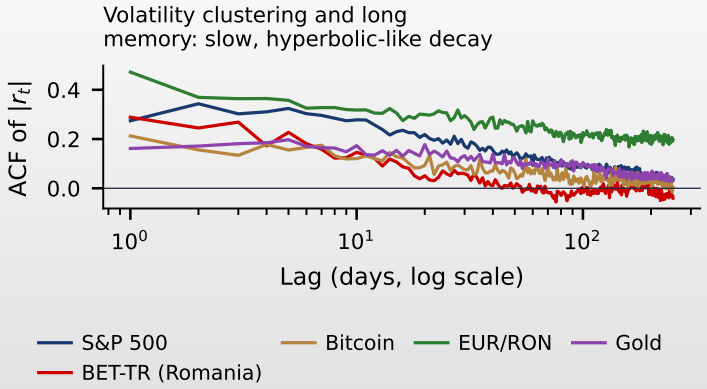
- Axes: abnormal volume $\tilde{v}_t$ (horizontal) and $|r_t|$ on a log scale (vertical); one point is one day
- $\hat{\rho}_S = 0.25$ for the S&P 500 (CI [0.22, 0.28]) and 0.39 for Bitcoin (CI [0.34, 0.44]): a positive, significant relation
- Mixture-of-distributions hypothesis Clark (1973): information flow drives both price moves and volume (survey of the evidence: Karpoff (1987))
- Bitcoin volume is exchange-reported and partly **wash trading** (fictitious trades in which buyer and seller coincide) Cong, Li, Tang & Yang (2023)

MFM_ch1_leverage_volume

Fact 8: Exponential or Hyperbolic Decay?

- **The question**: how fast does ρ_k, the autocorrelation of $|r_t|$ at lag k, decay as k grows?
- **Short memory**: $\sum_k |\rho_k| < \infty$
  - e.g. exponential decay $\rho_k \approx c\,\phi^k$, with $c > 0$ and $0 < \phi < 1$ (GARCH(1,1), Chapter 5)
- **Long memory** Ding, Granger & Engle (1993): hyperbolic decay $\rho_k \approx c\,k^{2d-1}$, with $0 < d < \frac{1}{2}$
  - d: the memory parameter; a larger d means slower decay, and $\sum_k \rho_k = \infty$
- **The test**: estimate d and test $H_0 : d = 0$ (short memory) with the semiparametric estimators of the next slides

Fact 8: Slow Decay of the ACF of $|r_t|$



MFM_ch1_volatility_clustering

Fact 8: the ACF of $|r_t|$ Decays Slowly in Every Market

- The chart: the ACF of $|r_t|$ (vertical) against the lag (horizontal, log scale), lags 1–250
- S&P 500: 0.27 at lag 1, 0.22 at lag 20, still 0.09 at lag 100
- BET-TR decays fastest
- Slow empirical decay is not yet established long memory
  - structural breaks, i.e. abrupt shifts in the level of volatility, can imitate it

MFM_ch1_volatility_clustering

Estimating the Memory Parameter d

- Long memory in the frequency domain: the spectral density grows without bound at low frequencies

$$f(\lambda) \approx G\lambda^{-2d} \quad \text{as } \lambda \to 0$$

  - $f(\lambda)$: the spectral density, the contribution of cycles of frequency λ to the variance of the series; small λ: long cycles
  - $G > 0$: a constant; semiparametric estimators use only the m lowest Fourier frequencies $\lambda_j = 2\pi j/T$, $j = 1, \ldots, m$

- Two estimators (formulas in the Appendix)
  - local Whittle Robinson (1995): $\sqrt{m}(\hat{d} - d) \to N(0, \frac{1}{4})$, so SE $= 1/(2\sqrt{m})$
  - log-periodogram regression Geweke & Porter-Hudak (1983): SE $= \pi/\sqrt{24m}$
- The choice $m = T^{0.65}$: a small m inflates the SE, a large m takes in frequencies where short-run (GARCH) dynamics bias $\hat{d}$

$\hat{d} \approx 0.35$–0.50 in All Five Markets

$\lvert r_t \rvert$, $m = T^{0.65}$	S&P 500	BET-TR	Bitcoin	EUR/RON	Gold
m	378	181	232	265	386
Local Whittle $\hat{d}$ (SE)	0.48 (0.03)	0.37 (0.04)	0.35 (0.03)	0.39 (0.03)	0.40 (0.03)
Log-periodogram $\hat{d}$ (SE)	0.50 (0.03)	0.41 (0.05)	0.35 (0.04)	0.38 (0.04)	0.42 (0.03)

- $\hat{d} \approx 0.35$–0.50 everywhere
  - at least 8 SEs from $d = 0$ (short memory) with both estimators
- The S&P 500 is close to the stationarity boundary $\frac{1}{2}$
  - for $d \geq \frac{1}{2}$ the series is no longer stationary

MFM_ch1_inference

Long Memory or Structural Breaks?

- Rare shifts in the level of volatility produce a slowly decaying ACF of $|r_t|$
  - and $\hat{d} > 0$ without long memory Diebold & Inoue (2001), Mikosch & Stărică (2004)
- Check: locate one variance break with the cumulative sum of squares Inclán & Tiao (1994)
  - then re-estimate d (local Whittle) on each side

$\lvert r_t \rvert$	Break date	Volatility before / after (%)	$\hat{d}$ before (SE)	$\hat{d}$ after (SE)
S&P 500	20 Jul 1998	12.6 / 19.3	0.28 (0.04)	0.51 (0.03)
BET-TR	17 Dec 2018	12.5 / 16.9	0.19 (0.05)	0.37 (0.04)
Bitcoin	19 Jun 2022	74.9 / 47.7	0.39 (0.04)	0.15 (0.05)
EUR/RON	27 Jun 2013	6.6 / 2.2	0.35 (0.04)	0.35 (0.04)
Gold	29 Nov 2005	13.2 / 18.3	0.30 (0.03)	0.37 (0.03)

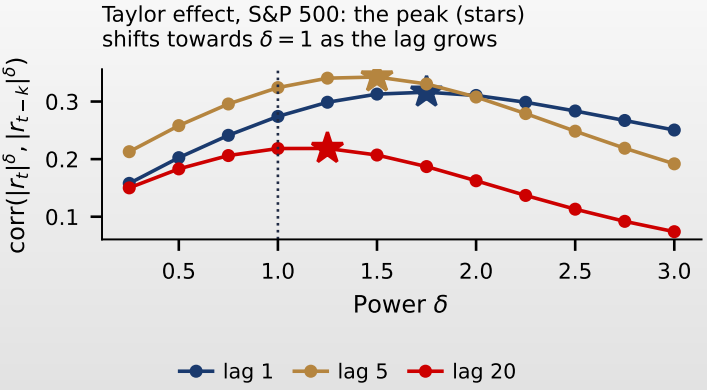
- $\hat{d}$ stays at least 3 SEs above 0 in every sub-sample (weakest: Bitcoin after 2022, $0.15/0.05 = 3$): one break does not explain the memory
- S&P 500 after 1998: $\hat{d} = 0.51$, above the stationarity boundary $\frac{1}{2}$ but within one SE of it; interpret with care
- $\hat{d}$ moves (Bitcoin $0.39 \to 0.15$): several breaks or a time-varying d remain open; the break dates are descriptive

MFM_ch1_inference

The Taylor Effect: the Question and the Statistic

- **The question**: which power of the size of returns is the most persistent?
- Statistic: $\rho_k(\delta) = \text{corr}(|r_t|^\delta, |r_{t-k}|^\delta)$, as a function of the power $\delta > 0$
  - $\delta = 1$: absolute returns; $\delta = 2$: squared returns
- **Taylor effect**: $\rho_k(\delta)$ peaks at a power δ close to 1, not at $\delta = 2$
  - documented by Taylor (1986), named by Granger & Ding (1995)
- The effect motivates power-GARCH models, which model σ_t^δ with an estimated δ (Chapter 5), and absolute-value volatility measures

The Taylor Effect



MFM_ch1_volatility_clustering

The Taylor Effect: Persistence Peaks below the Square

- The chart: $\rho_k(\delta)$ for lags 1, 5 and 20; stars mark the peak, the dotted line $\delta = 1$
- Our S&P 500 sample: maximum at $\delta = 1.75$ (lag 1), 1.5 (lag 5) and 1.25 (lag 20)
  - the peak is below 2 at every lag and moves towards 1 as the lag grows
  - at lag 1 the peak is closer to 2 than to 1: here the evidence for "$\delta \approx 1$" is mixed at short lags
- The location of the peak is a point estimate without a standard error: the comparison with $\delta = 1$ is descriptive

MFM_ch1_volatility_clustering

Which Inference Survives When Moments Are Missing

Statistic	Needs	With Hill $\hat{\alpha} \approx 2.3$–3.5 (Bitcoin: 4.2)
Mean, variance of r_t	$\alpha > 2$	finite, slow convergence
ACF of $\lvert r_t \rvert$, robust ACF bands, $\tilde{Q}$	$\alpha > 4$ for Normal limits	doubtful
Kurtosis, ACF of r_t^2, ARCH-LM, Q on r_t^2	$\alpha > 4$ to exist (and for ARCH-LM's χ^2 under i.i.d.), $\alpha > 8$ for Normal limits under GARCH-type dependence	no standard inference
Skewness	$\alpha > 6$ for a Normal limit	no standard inference
Quantiles, number of tail days, Hill $\hat{\alpha}$	no moments	valid, with dependence-robust SEs

- With an infinite fourth moment the sample ACF of r_t^2 converges slowly
  - its limit is not the Normal distribution Davis & Mikosch (1998)
  - the $\pm 1.96/\sqrt{T}$ band and the p-values of Q on r_t^2 and ARCH-LM are not reliable
- The conclusion "volatility clusters" is robust
  - the exact magnitudes are not robust
  - one reason behind the Taylor effect: $\lvert r_t \rvert$ needs fewer finite moments than r_t^2

Facts 7 and 11: Conditional Tails and Time Scales

Fact 7: conditional heavy tails

- Fit a volatility model, e.g. GARCH (Chapter 5)
  - and standardise: $\hat{z}_t = r_t / \hat{\sigma}_t$, with $\hat{\sigma}_t$ the model's volatility estimate for day t
- The kurtosis of $\hat{z}_t$ is lower than that of r_t, still above 3
- Models need heavy-tailed innovations (Student-t, possibly skewed)

Fact 11: asymmetry in time scales

- Weekly volatility predicts daily volatility better than the reverse
- Traders with different horizons react at different frequencies
- The heterogeneous autoregressive model of realised volatility (**HAR-RV**, Chapter 9)
  - realised volatility is the sum of squared intraday returns

- Both facts are documented in Cont (2001)
- Returns standardised by realised volatility from intraday data are close to the Normal distribution
  - Andersen, Bollerslev, Diebold & Ebens (2001) (Chapter 9)

Differences between the Five Markets

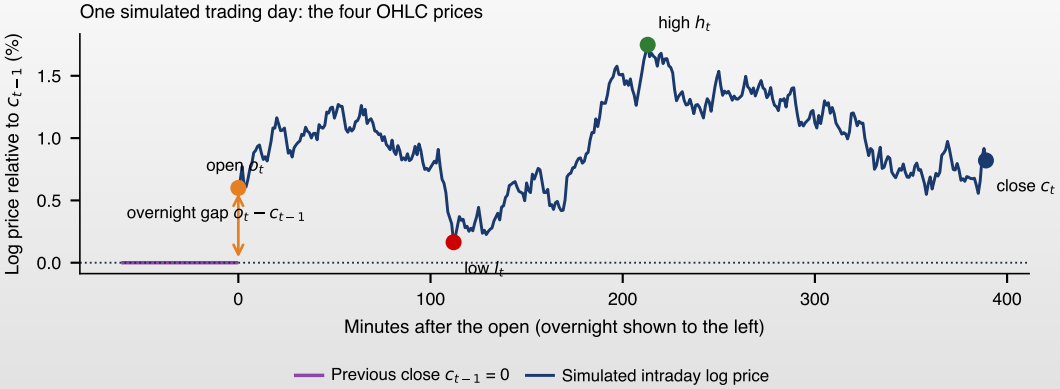
Market	Institutional feature	What differs	Candidate explanation
Bitcoin	trades 24/7; $A = 365$	volatility 66.6% versus 18.0% (S&P 500); worst day -46.5%; weak leverage (-0.07); ACF of $\lvert r_t \rvert$ at lag 100: 0.02 versus 0.09	no corporate leverage; volatility rises after large moves of either sign
EUR/RON	BNR daily fixing, managed float	volatility 4.4% but excess kurtosis 15.1; $\hat{\rho}_1 = +0.17$; largest moves October 2008 ($+2.9\%$, -2.5%)	adjustments smoothed over several days
BET-TR	total return, frontier/emerging market	skewness -1.40; worst day -11.9% on 19 Dec 2018 (bank-asset tax announced on 18 Dec, Government Emergency Ordinance 114/2018); $\hat{\rho}_1 = +0.06$	policy shocks; **non-synchronous trading** of constituents Campbell, Lo & MacKinlay (1997), ch. 3

- The explanations are hypotheses, not identified causes
- Low volatility is not low tail risk
- Heavy tails and volatility clustering hold in all five markets
  - autocorrelation and asymmetry depend on institutions

What the Day's High and Low Add

- **The question**: how precisely can we estimate a day's volatility if we also use the open, the high and the low, not only the close?
- **OHLC**: four prices of day t
  - O_t: the open; H_t: the high; L_t: the low; C_t: the close
- The next chart: one simulated day; the vertical axis is the log price relative to the previous close c_{t-1}, in %
  - the overnight gap $o_t - c_{t-1}$ sits left of the open
- **Range-based** estimators use the whole path of the day: the range H_t/L_t is large on volatile days, even if the close equals the open

Open, High, Low and Close (OHLC) Prices of One Day



MFM_ch1_seminar_tests

The Log Range and the Three Returns of a Day

- Logs of the four prices

$$o_t = \ln O_t, \qquad h_t = \ln H_t, \qquad l_t = \ln L_t, \qquad c_t = \ln C_t$$

  - the lower-case letter is the log of the price with the same initial

- Log range of day t

$$h_t - l_t = \ln(H_t/L_t) \geq 0$$

  - approximately the percentage gap between the day's high and low

- Three log returns of day t
  - close-to-close: $c_t - c_{t-1}$
  - overnight: $o_t - c_{t-1}$
  - intraday: $c_t - o_t$

The Close-to-Close Estimator and Efficiency

- **Close-to-close estimator**: the standard deviation of $c_t - c_{t-1}$
  - one observation per day
- **Efficiency** of an estimator X: $\text{Var}(\hat{\sigma}^2_{CC})/\text{Var}(\hat{\sigma}^2_X)$
  - an efficiency of 5 means the same precision with 5 times fewer days

The Parkinson Estimator: Variance from the Day's Range

- **Parkinson**, Parkinson (1980): estimates the daily variance from the daily range only

$$\hat{\sigma}_P^2 = \frac{1}{4 \ln 2} \frac{1}{n} \sum_{t=1}^{n} (h_t - l_t)^2$$

  - n: the number of days; $h_t - l_t$: the log range of day t
  - for a Brownian motion with daily variance σ^2, $\mathbb{E}[(h_t - l_t)^2] = 4 \ln 2 \, \sigma^2$; dividing by $4 \ln 2$ makes the estimator unbiased
  - $\hat{\sigma}_P^2$ is a daily variance; the annual volatility is $\sqrt{A \, \hat{\sigma}_P^2}$

- Assumption: a Brownian motion without drift and without overnight jumps
  - Brownian motion: the continuous-time random walk (Chapter 11)
  - efficiency relative to close-to-close: about 5 (Seminar 1, A8)

The Garman–Klass Estimator: the Range plus the Open-to-Close Move

- **Garman–Klass**, Garman & Klass (1980): combines the range with the open-to-close move

$$\hat{\sigma}^2_{GK} = \frac{1}{n}\sum_{t=1}^{n}\left[\tfrac{1}{2}(h_t - l_t)^2 - (2\ln 2 - 1)(c_t - o_t)^2\right]$$

  - $c_t - o_t$: the open-to-close log return
  - the weights $\frac{1}{2}$ and $2\ln 2 - 1 \approx 0.386$ come from the minimum-variance estimator derived by Garman and Klass
  - the same assumptions as Parkinson: no drift, no overnight jumps
- Efficiency relative to close-to-close: 7–8

The Rogers–Satchell Estimator: Unbiased under a Drift

- **Rogers–Satchell**, Rogers & Satchell (1991): unbiased under a non-zero **drift** (expected trend of the log price)

$$\hat{\sigma}^2_{RS} = \frac{1}{n}\sum_{t=1}^{n}[(h_t - o_t)(h_t - c_t) + (l_t - o_t)(l_t - c_t)]$$

  - the terms measure how far the high and the low moved away from the open and from the close
  - both products are ≥ 0, because $l_t \leq o_t, c_t \leq h_t$
  - for a Brownian motion with drift their mean is exactly the daily variance
  - it does not include the overnight jump

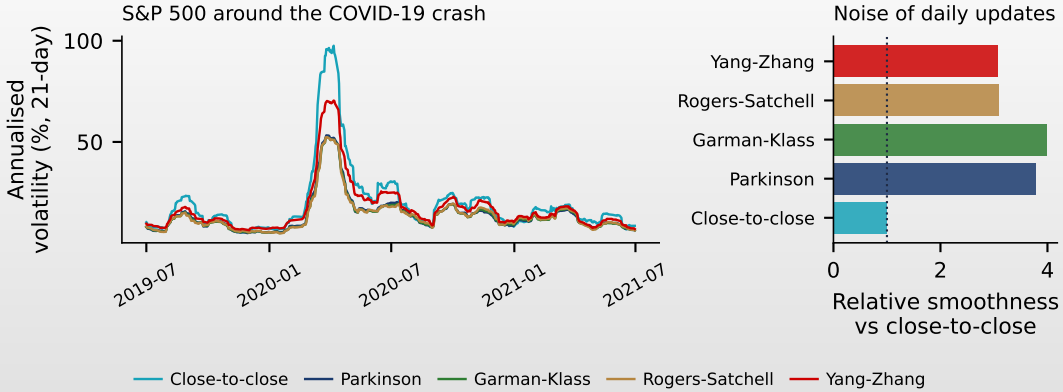
The Yang–Zhang Estimator: Adding the Overnight Jump

- **Yang–Zhang**, Yang & Zhang (2000): adds the overnight (close-to-open) jump

$$\hat{\sigma}^2_{YZ} = \hat{\sigma}^2_{\text{overnight}} + k\,\hat{\sigma}^2_{\text{open-close}} + (1-k)\,\hat{\sigma}^2_{RS}, \quad k = \frac{0.34}{1.34 + \frac{n+1}{n-1}}$$

  - $\hat{\sigma}^2_{\text{overnight}}$: the sample variance of $o_t - c_{t-1}$, the overnight return
  - $\hat{\sigma}^2_{\text{open-close}}$: the sample variance of $c_t - o_t$
  - k: the weight, chosen to make the combination minimum-variance (not the Hill k); $k \approx 0.145$ for large n
  - minimum-variance combination among drift-independent estimators robust to opening jumps
- Range estimators are the bridge to realised volatility from intraday data (Chapter 9)

Volatility Estimators on the S&P 500



MFM_ch1_vol_estimators

Missing the Overnight Jumps, Range Estimators Read Lower

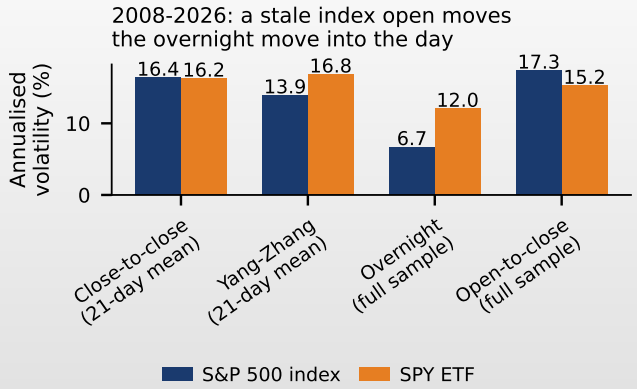
- 21-day windows, 2008–2026; averages
  - close-to-close 16.4%, Yang–Zhang 13.9%, Parkinson 13.0%, Garman–Klass 12.4%, Rogers–Satchell 12.3%
- Pure range estimators are lower: they miss overnight gaps
- Right panel, "relative smoothness" of estimator X

$$\left[\frac{\text{std. dev. of daily changes, close-to-close estimate}}{\text{std. dev. of daily changes, estimate } X}\right]^2$$

  - both estimates use 21-day windows
  - equal to 1 for close-to-close; larger values mean the estimate jumps less from one day to the next
- The range-based series are 3–4 times smoother on this measure
  - a property of the estimates, not by itself proof of efficiency: it does not compare them with the true σ

MFM_ch1_vol_estimators

Stale Opens: the S&P 500 Index versus SPY, an Exchange-Traded Fund (ETF)



MFM_ch1_seminar_tests

Stale Opens: Why the Index Open Lags Behind

- **Stale open**: the index open is computed from the constituents' first prices
  - at 9:30 many stocks have not traded yet
  - so their previous close enters the open
- SPY, the ETF that tracks the S&P 500, opens at a traded price
- Open = previous close
  - about 96% of days in 1996–2005 and still 16% in 2007
  - 3.5% over 2008–2026 (up to 26% in 2012–2013)
  - range-based estimators therefore use data from 2008 on

Stale Opens: the Effect on Volatility Estimators

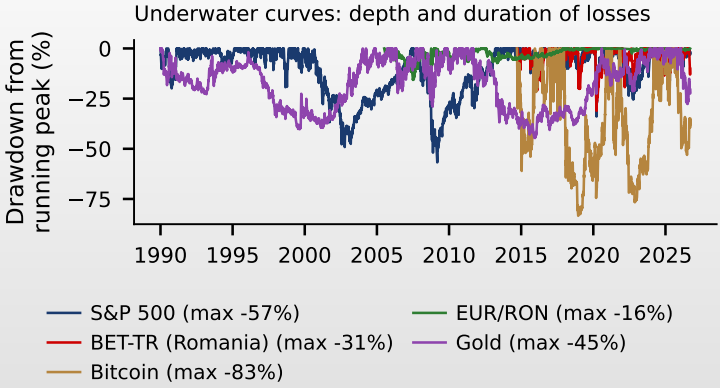
- A stale open moves part of the overnight move into the day
  - index overnight volatility: 6.7%
  - SPY, whose open is a traded price: 12.0%
- Yang–Zhang falls below close-to-close on the index only
  - use a traded instrument for estimators that use the open (Seminar 1, B15)

MFM_ch1_seminar_tests

Drawdowns: the Question, the Statistic and the Benchmark

- **The question**: is the deepest observed fall larger than a random walk with the same drift and volatility would produce?
- Drawdown $DD_t = P_t / \max_{s \le t} P_s - 1 \in [-1, 0]$ (Chapter 0); maximum drawdown $MDD = \min_t DD_t$
- On the log scale: $D = \max_t \left(\max_{s \le t} \ln P_s - \ln P_t \right) = -\ln(1 + MDD) \ge 0$
- The benchmark Magdon-Ismail, Atiya, Pratap & Abu-Mostafa (2004): $\ln P_t$ is a Gaussian random walk with annual drift μ and annual volatility σ, over Y years
  - $\mu = 0$: $\mathbb{E}[D] = \sqrt{\pi/2}\,\sigma\sqrt{Y}$, growing in proportion to $\sqrt{Y}$
  - $\mu > 0$: $\mathbb{E}[D] \approx \frac{2\sigma^2}{\mu}\left[\frac{1}{4}\ln\frac{\mu^2 Y}{2\sigma^2} + 0.49\right]$, growing only like $\ln Y$ (an approximation for large $\mu^2 Y/(2\sigma^2)$)
- **The test**: p-value $= P(D \ge D_{\text{obs}})$ under the benchmark, from 5,000 paths simulated with the estimated drift and volatility

Drawdowns Across Markets



MFM_ch1_risk_measures

The Maximum Drawdown: How Far a Price Can Fall from a Peak

- The chart: underwater curves $100\,DD_t$ (%); the legend gives each market's MDD
- MDD: Bitcoin -83%, S&P 500 -57%, gold -45%, BET-TR -31% (March 2020), EUR/RON -16%
- S&P 500: 9 October 2007 (1,565.15) to 9 March 2009 (676.53): -56.8%
- **Recovery**: the price regaining its previous peak
  - after a fall of x it needs a gain of $1/(1-x)-1$; here $1/(1-0.568)-1=+131\%$

MFM_ch1_risk_measures

The Maximum Drawdown against a Gaussian Random Walk

	S&P 500	BET-TR	Bitcoin	EUR/RON	Gold
Observed D (MDD)	0.84 (−57%)	0.37 (−31%)	1.80 (−83%)	0.17 (−16%)	0.59 (−45%)
$\mathbb{E}[D]$, $\mu = 0$	1.37	0.67	2.89	0.26	1.24
$\mathbb{E}[D]$, $\mu = \hat{\mu}$: formula / simulation	0.65 / 0.66	0.26 / 0.26	1.48 / 1.55	0.14 / 0.15	0.62 / 0.64
p-value, $P(D \geq D_{\text{obs}})$	0.17	0.07	0.26	0.29	0.52

- The driftless benchmark overstates $\mathbb{E}[D]$ about twofold: a positive drift limits how deep prices fall
- With the estimated drift no MDD is unusual: p-values from 0.07 (BET-TR) to 0.52 (gold)
- The benchmark ignores heavy tails and volatility clustering, so it is only a first point of comparison
- The MDD grows with the sample length: compare only equal horizons (Appendix)

generate_all_charts

Recovery Duration after the Maximum Drawdown

- **Recovery duration**: the time from the peak until the price reaches it again (recovery: the price regaining its previous peak)

	MDD	Peak	Trough	Peak regained
S&P 500	−57%	Oct 2007	Mar 2009	Mar 2013 (5.5 years)
BET-TR	−31%	Jan 2020	Mar 2020	Dec 2020 (0.9 years)
Bitcoin	−83%	Dec 2017	Dec 2018	Nov 2020 (3.0 years)
EUR/RON	−16%	Dec 2005	Jul 2007	Jan 2008 (2.0 years)
Gold	−45%	Sep 2011	Dec 2015	Jul 2020 (8.9 years)

- Depth and duration are different risks
  - gold's drawdown was shallower than Bitcoin's but lasted three times longer
- Fast V-shaped recoveries (BET-TR 2020)
  - versus long underwater periods (gold 2011–2020)
- For EUR/RON a "drawdown" is a fall of the euro against the leu

MFM_ch1_risk_measures

The Sharpe Ratio: Excess Return per Unit of Volatility

- **Sharpe ratio** Sharpe (1966), Sharpe (1994)
  - the excess return earned per unit of volatility

$$SR = \frac{\mu_{\text{ann}} - r_f}{\sigma_{\text{ann}}}$$

  - r_f: the risk-free rate; $r_f = 0$ here, for comparability
  - $\mu_{\text{ann}} = A\bar{R}$: the annual mean of **simple** returns
  - $\bar{R}$: the mean daily simple return
  - σ_{ann}: the annual volatility of simple returns
  - a higher SR means more return for the same risk
- S&P 500 1990–2026: $\mu_{\text{ann}} = 9.95\%$, $\sigma_{\text{ann}} = 17.97\%$, $SR = 0.55$
  - μ_{ann} = mean log return 8.33% plus 1.62 pp, close to $\sigma^2/2 = 1.61$ pp
  - this is not the log-return mean $A\bar{r}$ of the annualisation slide: with 8.33% one would get $8.33/17.97 = 0.46$

The Sortino Ratio: Only Losses in the Denominator

- **Sortino ratio** Sortino & van der Meer (1991)
  - the same numerator as the Sharpe ratio
  - in the denominator: only the volatility of losses

$$SoR = \frac{\mu_{\text{ann}} - r_f}{\sigma^-_{\text{ann}}}, \qquad \sigma^- = \sqrt{\frac{1}{T}\sum_{t=1}^{T} \min(R_t, 0)^2}$$

  - σ^-: the daily downside deviation (in the seminar: the downside semi-deviation $\hat{\sigma}_D$)
  - $\min(R_t, 0)$ keeps only negative returns
  - the average runs over all T days; positive days count as 0
  - $\sigma^-_{\text{ann}} = \sqrt{A}\,\sigma^-$: the annualised value
- S&P 500: $\sigma^-_{\text{ann}} = 12.68\%$, $SoR = 0.78$

The Calmar Ratio: Growth per Unit of Maximum Drawdown

- **Calmar ratio**
  - the annual growth earned per unit of maximum drawdown

$$\text{Calmar} = \frac{\text{CAGR}}{|MDD|}$$

  - CAGR: the compound annual growth rate
  - MDD: the maximum drawdown in the sample
- S&P 500: $8.69\%/56.78\% = 0.15$
  - the value is driven by one episode
- The Calmar ratio has no usable sampling theory
  - its maximum drawdown grows with the sample length

Sharpe Ratios with Standard Errors

- Delta method, i.i.d. returns from the Normal distribution Lo (2002)
  - SR_d: the daily Sharpe ratio; $\widehat{SR}_d$: its estimate; $\mathrm{Var}(\widehat{SR}_d) \approx (1 + SR_d^2/2)/T$
  - annualised, $SR_{\text{ann}} = \sqrt{A}\, SR_d$: $\mathrm{SE}(\widehat{SR}_{\text{ann}}) \approx \sqrt{A(1 + SR_d^2/2)/T} \approx 1/\sqrt{Y}$, with $Y = T/A$ years
  - the span matters, not the frequency

	S&P 500	BET-TR	Bitcoin	EUR/RON	Gold
Years Y	36.7	12.0	12.0	21.2	36.7
Sharpe ratio (SE, Normal)	0.55 (0.17)	1.34 (0.29)	0.98 (0.29)	0.43 (0.22)	0.48 (0.17)

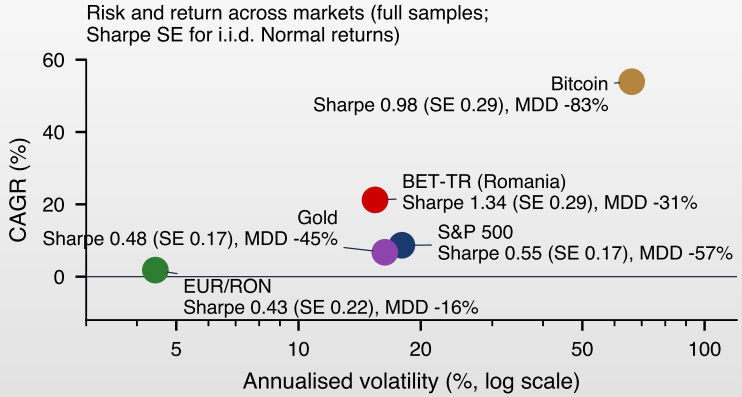
MFM_ch1_risk_measures

BET-TR versus S&P 500: the Sharpe Difference Is Not Significant at 5%

- Clustering: HAC or studentised block-bootstrap tests Ledoit & Wolf (2008)
  - they still need finite fourth moments, which the Hill estimates put in doubt
- BET-TR versus S&P 500, common days 2014–2026
  - difference 0.61, $z = 1.75$ ($p = 0.08$)
  - z: the difference divided by its standard error, compared with $N(0,1)$
  - block-bootstrap 95% CI $[-0.01, 1.21]$ (Seminar 1, B14)

MFM_ch1_risk_measures

Risk and Return Across Markets



MFM_ch1_risk_measures

Why the Risk–Return Chart Is Not a Ranking

- Axes: annualised volatility (horizontal, log scale) and CAGR (vertical)
- Labels: the Sharpe ratio, its standard error (SE) and the MDD
  - SE: the formula of Sharpe ratios with standard errors, valid for i.i.d. Normal simple returns; full table in the Appendix
- Different sample periods: not a ranking
- S&P 500 without dividends; BET-TR with dividends and in RON, and RON short-term rates were well above USD rates: with $r_f = 0$ the BET-TR Sharpe ratio is inflated
- Heavy tails and negative skewness flatter the Sharpe ratio: report it with drawdowns, the 1% VaR and the tail index $\hat{\alpha}$

MFM_ch1_risk_measures

Case Study (1/11): How Much of Volatility Do Past Returns Alone Explain?

- Paper: Guyon & Lekeufack (2023), *Quantitative Finance*
  - the paper builds on Bollerslev, Hood, Huss & Pedersen (2018)
- Research question
  - what share of an equity index's volatility is explained by its past daily returns alone?
  - what functional form does this dependence take?
- Three dependent variables, modelled separately (Sections 3.4–3.5 of the paper)
  - VIX (the Cboe volatility index): the implied volatility of the S&P 500 over the next 30 days, computed from option prices
  - **implied volatility**: the volatility that makes an option-pricing model match observed option prices, i.e. the market's forward-looking volatility (Chapter 12)
  - VIX9D: the same measure over a 9-day horizon
  - next-day realised volatility, computed from 5-minute returns

Case Study (2/11): One Model for Volatility Clustering, Long Memory and Leverage

- Requirement: a single model must reproduce simultaneously
  - volatility clustering (Fact 6)
  - the slow decay of the autocorrelation of $|r_t|$ (Fact 8)
  - the leverage effect (Fact 9)
- Why it matters: if volatility is mostly a function of the price path, it is mostly endogenous
  - a **path-dependent volatility (PDV)** model needs little extra randomness
  - unlike stochastic volatility models (Chapters 11–12)
- Data: daily closes of the S&P 500 (US), Euro Stoxx 50 (euro area), FTSE 100 (UK), DAX (Germany) and Nikkei 225 (Japan) indices
  - training 1 Jan 2000 – 31 Dec 2018
  - test 1 Jan 2019 – 15 May 2022

Case Study (3/11): Volatility as a Function of the Trend and of Recent Volatility

- Equation (3.7) in the paper: today's volatility is a linear function of two summaries of past returns

$$\sigma_t = \beta_0 + \beta_1 R_{1,t} + \beta_2 \Sigma_t$$

- Notation
  - σ_t: the dependent variable, for example the VIX at the close of day t
  - S_{t_i}: the index level at the close of day t_i; $t_i \leq t$ runs over the days up to and including t
  - $r_{t_i} = S_{t_i}/S_{t_{i-1}} - 1$: the **simple** return of day t_i (the paper's notation; elsewhere in this chapter r_t is the log return)
  - $R_{1,t}$: the trend component, built from past returns
  - Σ_t: the volatility component, built from past squared returns
  - $\beta_0, \beta_1, \beta_2$: coefficients estimated from the data

Case Study (4/11): How Much Does an Old Day Count?

- A **kernel** $K(\tau)$ gives the weight of a day of age τ, with $\tau = i\Delta t$ for a day i days old and $\Delta t = 1/252$
- The daily weights $K(i\Delta t)\Delta t$ are positive and sum to 1 (the vertical axis of the kernel chart)
- The paper uses a **time-shifted power law (TSPL)**, eq. (3.9)

$$K(\tau) = \frac{(\tau + \delta)^{-\alpha}}{Z_{\alpha,\delta}}, \qquad \alpha > 1,\ \delta > 0,\ \tau = i\Delta t,\ i = 0, 1, \ldots, 999$$

  - $Z_{\alpha,\delta}$: the normalising constant over the last 1,000 days
  - α: how fast the weights decay; a small α means long memory
  - δ: the time shift, in years; a small δ concentrates the weight on the last few days ($\delta = 0.05$ years ≈ 12.6 days)
- Resulting shape: large weight on the last few days (short memory) and a slowly decaying power-law tail (long memory, Fact 8)
- The model uses two kernels: K_1 for the trend and K_2 for volatility, with parameters (α_1, δ_1) and (α_2, δ_2)
- Here α, δ and K are not the tail index, the Taylor-effect power or the kurtosis

Case Study (5/11): The Recent Trend and a Weighted Historical Volatility

- Trend component: a weighted sum of past returns

$$R_{1,t} = \sum_{t_i \le t} K_1(t - t_i)\, r_{t_i}$$

  - $t - t_i$: the age of day t_i, measured in years as in the paper (one day $= 1/252$ years); $K_1(t - t_i)$: the weight of that day
  - $R_{1,t} < 0$ after a run of falls, $R_{1,t} > 0$ after a run of rises
- Volatility component: the square root of a weighted sum of squared returns

$$\Sigma_t = \Big(\sum_{t_i \le t} K_2(t - t_i)\, r_{t_i}^2 \Big)^{1/2}$$

  - The weights $K_j(i\Delta t)\Delta t$ sum to 1: $R_{1,t}$ and Σ_t are weighted means, annualised with 252 and $\sqrt{252}$
- Interpreting the coefficients
  - $\beta_1 < 0$: volatility rises after falls ($R_{1,t} < 0$), i.e. the leverage effect
  - $\beta_2 > 0$: high recent volatility raises today's volatility, i.e. volatility clustering, as in GARCH
  - $\beta_2 < 1$ is an empirical finding, not a constraint ($\hat{\beta}_2 = 0.87$)
  - the VIX moves less than one-for-one with Σ_t
  - $\beta_0 > 0$ supplies part of the level

Case Study (6/11): Seven Parameters, Scored Also out of Sample

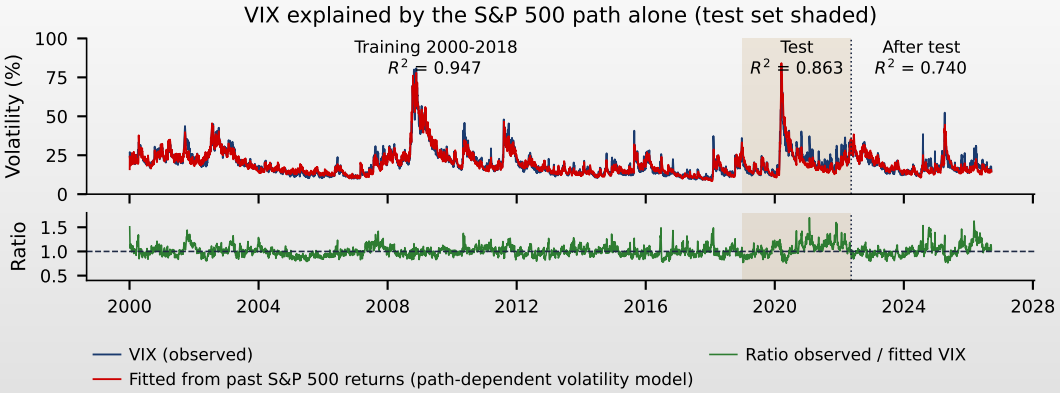
- Seven parameters: $\beta_0, \beta_1, \beta_2, \alpha_1, \delta_1, \alpha_2, \delta_2$
  - estimated by nonlinear least squares on the training set: minimise $\sum_t (\sigma_t - \hat{\sigma}_t)^2$
  - $\hat{\sigma}_t$: the model value for day t
  - estimation details: Appendix
- Scores, computed separately on the training and the test set
  - RMSE (root mean squared error) $= \sqrt{\frac{1}{T}\sum_t (\sigma_t - \hat{\sigma}_t)^2}$: the typical error, in volatility units (decimals: 0.01 = 1 VIX point)
  - $R^2 = 1 - \sum_t (\sigma_t - \hat{\sigma}_t)^2 / \sum_t (\sigma_t - \bar{\sigma})^2$: the share of the variation of σ_t explained by the model
  - $\bar{\sigma}$: the mean of σ_t; $R^2 = 1$ is a perfect fit, $R^2 = 0$ is no better than the mean

Case Study (7/11): Past Returns Explain 80–92% of Implied Volatility

Index	Implied volatility (paper, Table 2) R^2 train	R^2 test	Realised volatility (paper, Table 4) R^2 train	R^2 test
S&P 500 (VIX)	0.946	0.855	0.738	0.654
S&P 500 (VIX), our replication	0.947	0.863	–	–
S&P 500 (9-day VIX)	0.876	0.914	–	–
Euro Stoxx 50	0.929	0.913	0.672	0.682
Nikkei 225	0.890	0.800	0.563	0.504

- Past index returns alone explain 80–92% of the variance of implied volatility out of sample
  - FTSE 100 and DAX in the Appendix
- On our data (S&P 500 and VIX): RMSE 0.020 (2 VIX points) training and 0.034 test
- Realised volatility scores are lower
  - 5-minute realised volatility is a noisy target (Remark 3 in the paper)

Case Study (8/11): Observed VIX and VIX Fitted from S&P 500 Returns



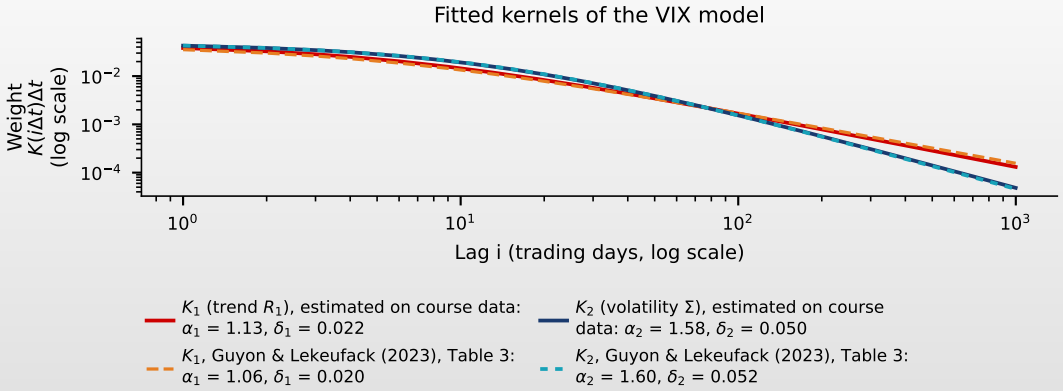
MFM_ch1_path_dependent_vol

Case Study (9/11): After 2022, a Smaller Error but Also a Lower R^2

- Fitted on 3 Jan 2000 – 31 Dec 2018 (4,779 days)
  - then parameters frozen
- The test set is shaded
- After the test set, 16 May 2022 – 18 Sep 2026: $R^2 = 0.740$, RMSE 0.026
  - a lower RMSE, but the VIX varied less (standard deviation 5.0 versus 9.3 points)
  - so R^2 falls
- Ratio observed/fitted
  - mean 0.85 in April–May 2020
  - mean 1.16 from October 2020 to May 2022
  - 8 April 2025: VIX 52.3 versus 39.2 fitted

MFM_ch1_path_dependent_vol

Case Study (10/11): Fitted Kernels



MFM_ch1_path_dependent_vol

Case Study (11/11): One Kernel for Short and Long Memory; What Stays Exogenous

- Daily weights $K(i\Delta t)\Delta t$, log–log scale; in parentheses, Table 3 of the paper
  - K_1 (trend): $\hat{\alpha}_1 = 1.13$ (1.06), $\hat{\delta}_1 = 0.022$ (0.020)
  - K_2 (volatility): $\hat{\alpha}_2 = 1.58$ (1.60), $\hat{\delta}_2 = 0.050$ (0.052)
- The last 5 days carry 20% of the weight of K_2
  - the days beyond lag 250 still carry 10%
  - short and long memory in one kernel
- Limits stated by the authors (Section 5 of the paper)
  - a small exogenous part remains
  - the ratio of observed to fitted VIX mean-reverts around 1
  - this ratio needs a stochastic factor
- Open extension (project seed)
  - does the same formula describe the realised volatility of Bitcoin or of BET-TR?

MFM_ch1_path_dependent_vol

From Evidence to Model Requirements

Evidence (five markets)	Model requirement	Where in this course
quantile (Crow–Siddiqui) excess kurtosis > 0 in all five markets; Hill $\hat{\alpha} \approx 2.3$–3.5 (Bitcoin: 4.2)	heavy-tailed innovations, also conditionally	Ch. 5, Ch. 7 (extreme value theory)
ACF of $\lvert r_t \rvert$ positive for months; $\hat{d} \approx 0.35$–0.50	persistent, time-varying volatility; long memory or breaks	Ch. 5, Ch. 9
$L(1) = -0.12$ (S&P 500, BET-TR); reversed for EUR/RON	asymmetric response to bad news where it exists	Ch. 5 (GJR-GARCH, EGARCH)
$\hat{\rho}_1$ significant (robust SEs) only for the S&P 500 and EUR/RON; economically small in all markets	approximately uncorrelated returns; market microstructure matters	Ch. 2, Ch. 10
quantile skewness negative only for the S&P 500	asymmetry is market-specific	Ch. 7–8

- The random walk with i.i.d. increments from the Normal distribution (Bachelier–Black–Scholes world) violates Facts 2, 3 and 5–11
  - Facts 1 and 4 hold trivially
- ARCH-LM significance supports volatility clustering descriptively
  - with doubtful fourth moments it is not a formal proof

Open Question: Are Tail Indices Stable over Time?

- The tail index α is usually treated as a constant
- Crypto-assets changed in structure
  - futures (2017), spot Bitcoin ETFs (January 2024), institutional holders
  - Liu & Tsyvinski (2021): crypto returns are not explained by the risk factors of stocks, currencies or precious metals
- Hypothesis proposed by an AI assistant
  - after the launch of the spot ETFs (January 2024), the entry of institutional investors has made the tails of Bitcoin's return distribution thinner
  - i.e. it has raised the tail index α

The Discovery Loop for Bitcoin's Tail Index

- **Literature**: ask for peer-reviewed work with DOIs, then resolve each DOI yourself
  - Hoga (2017): change-point tests for the tail index valid under β-mixing dependence
  - β-mixing: dependence that vanishes with the distance in time, e.g. in GARCH models
- **Competing hypotheses**
  - a change in tail shape
  - a fall in volatility only (scale does not move α)
  - sampling noise
- **Data and code**: replicate the chapter's Hill estimates first (Bitcoin 4.2, S&P 500 3.1)
- Fixed in advance
  - $k = 2.5\%$ of n, the break date 11 January 2024
  - block-bootstrap SEs, a placebo asset

Mini-Case: Did Bitcoin's Tail Get Thinner after the ETFs?

- Test: Hill estimate for $|r_t|$, $k = 2.5\%$ of n
  - $H_0 : \alpha_{\text{post}} = \alpha_{\text{pre}}$ against $H_1 : \alpha_{\text{post}} > \alpha_{\text{pre}}$
  - $\hat{\alpha}_{\text{pre}}$, $\hat{\alpha}_{\text{post}}$: the Hill estimates in the two subperiods; s: their standard errors, from a moving-block bootstrap (blocks of 20 days)
  - $z = (\hat{\alpha}_{\text{post}} - \hat{\alpha}_{\text{pre}})/\sqrt{s_{\text{pre}}^2 + s_{\text{post}}^2}$: the difference divided by its standard error
  - under H_0, $z \approx N(0,1)$; reject H_0 at 5% if $z > 1.645$
- Results
  - before (Sep 2014 – 10 Jan 2024, $k = 85$): $\hat{\alpha} = 4.16$ ($s = 0.52$)
  - after (11 Jan 2024 – Sep 2026, $k = 24$): $\hat{\alpha} = 4.31$ ($s = 0.91$)
  - $z = 0.15$
- Placebo, S&P 500, 17 Sep 2014–18 Sep 2026; break 11 Jan 2024, $k = 58/16$ before/after: $2.82 \to 2.86$, $z = 0.03$

MFM_ch1_ai_discovery

A Non-Significant Result with Low Power: What We Cannot Conclude

- **Minimum detectable change**, one-sided 5% test with 80% power
  - $(1.645 + 0.842) \times 1.05 = 2.6$
  - 1.645: the one-sided 5% critical value; 0.842: the 80% quantile of $N(0, 1)$, for 80% power
  - $1.05 = \sqrt{0.52^2 + 0.91^2}$ is the SE of the difference
  - the Normal-distribution approximation is rough with 24 tail observations
  - the data cannot distinguish an unchanged tail from a large thinning of the tail
- A non-significant result does not show the absence of an effect
  - wait for more data or pool assets

MFM_ch1_ai_discovery

Project Seed

- **Question**: did the tail index of Bitcoin and Ethereum change with institutionalisation (futures 2017, spot ETFs 2024)?
  - raw and GARCH-standardised returns (shape versus scale)
  - loss and gain tails separately
  - Bitcoin uses the course series; Ethereum is not in the course data: document its source, the close time and time zone, and the currency (USD) before estimating
- **Replicate first**: the chapter's Hill table and the mini-case numbers
- **Pre-register**, committed before the results
  - H_0, H_1, the rule for k, break dates and bootstrap
- **Report**: the change-point test of Hoga (2017) as a Quantlet
  - estimates, both SEs, the minimum detectable change
- **Log**: prompts in `AI_USE.md`
  - every caught error (an i.i.d. SE, a date chosen from the data) in `AI_ERRORS.md`
  - reusable prompts (▸ Appendix) and checks (▸ Appendix)

What We Keep: the Size of Returns Is Forecastable, the Direction Hardly

Takeaways

- Returns are uncorrelated but not independent: their size is predictable
- Heavy tails are the rule: tail indices near 3
  - the fourth moment very likely does not exist
  - kurtosis-based inference fails
- Volatility is persistent (long memory or breaks)
  - in equity markets it reacts asymmetrically to losses
- Every test needs a reference distribution that the data may violate

Questions for the room

- Why does a mixture with σ_t drawn independently each day have heavy tails but no volatility clustering?
- Why does the S&P 500 statistic fall from $Q(10) = 91.6$ to $\tilde{Q}(10) = 19.0$ with robust standard errors?
- Why can the tail index be estimated with the Hill estimator when the kurtosis does not exist?

- Next chapter: are returns predictable at all? (Chapter 2)

Formula Sheet with Assumptions

Returns and annualisation

- $R_t = P_t/P_{t-\mathbf{1}} - 1$, $r_t = \ln(1+R_t)$, $r_t(k) = \sum_{j=\mathbf{0}}^{k-\mathbf{1}} r_{t-j}$, $R_{p,t} = \sum_i w_i R_{i,t}$
- $\text{CAGR} = (P_T/P_{\mathbf{0}})^{\mathbf{1}/Y} - 1$
- $\sigma_{\mathbf{ann}} = \sqrt{A}\,\hat{\sigma}$: uncorrelated returns, constant variance

Tests

- $JB = T\left[\hat{S}^{\mathbf{2}}/6 + (\hat{K}-3)^{\mathbf{2}}/24\right]$: $\chi^{\mathbf{2}}_{\mathbf{2}}$ under i.i.d. Normal
- $Q(m) = T(T+2)\sum_{k=\mathbf{1}}^{m} \hat{\rho}_k^{\mathbf{2}}/(T-k)$: $\chi^{\mathbf{2}}_m$ under i.i.d.
- $\tilde{Q}(m) = \sum_k \hat{\rho}_k^{\mathbf{2}}/\hat{\tau}_k$: $\mathbb{E}r^{\mathbf{4}} < \infty$, zero cross terms
- Hill $\text{SE}(\hat{\alpha}) \approx \hat{\alpha}/\sqrt{k}$: independent Pareto tail

Volatility

- Parkinson $\frac{\mathbf{1}}{\mathbf{4}\ln \mathbf{2}}\overline{(h-l)^{\mathbf{2}}}$, the bar a mean over days: driftless, no overnight jumps
- Mixture $K = 3\,\mathbb{E}[\sigma^{\mathbf{4}}]/\mathbb{E}[\sigma^{\mathbf{2}}]^{\mathbf{2}}$: σ_t independent of ε_t

Performance

- $SR = (\mu - r_f)/\sigma$ on simple returns: $\mu = A\bar{R}$, not the log mean $A\bar{r}$
  - $\ln(1+\text{CAGR}) \approx \mu - \sigma^{\mathbf{2}}/2$ (log-scale volatility drag)
  - $\text{SE} \approx 1/\sqrt{Y}$ for i.i.d. Normal returns
- $SoR = (\mu - r_f)/\sigma^{-}$
- $DD_t = P_t/\max_{s\le t} P_s - 1$; Calmar $= \text{CAGR}/|MDD|$

Back

Prices Before Computers, the Stock Ticker



Stock ticker designed by Thomas Edison

(Smithsonian collection)

Photo: Jaclyn Nash, Smithsonian National Museum of American History; public domain; Wikimedia Commons

- Before computers, trades reached brokers as prices printed on a paper **ticker tape**
- A daily close is one snapshot of this stream of prices
  - returns are built from such snapshots
- The choice of snapshot (close, fixing, a midnight snapshot) matters
  - as Chapter 0 showed
- Prices at the level of individual trades: market microstructure, Chapter 10

Back

The People Behind the Stylised Facts (1/2)



Carl Friedrich Gauss (1777–1855)

Painting: C.A. Jensen (1840); public domain; Wikimedia Commons

- The **Normal distribution**: the benchmark that returns violate



Louis Bachelier (1870–1946)

Photo: unknown author; public domain; Wikimedia Commons

- Prices as a random walk, in his thesis Bachelier (1900)



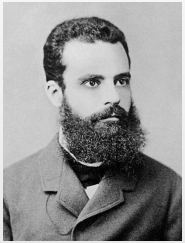
William S. Gosset, "Student" (1876–1937)

Photo: unknown author; public domain; Wikimedia Commons

- The Student-t distribution Student (1908), today a model for heavy tails

Back

The People Behind the Stylised Facts (2/2)



Vilfredo Pareto (1848–1923)

Photo: unknown author (1870s); public domain; Wikimedia Commons



Benoît Mandelbrot (1924–2010)

Photo: Rama (2007); CC BY-SA 2.0 France; Wikimedia Commons

- Pareto: **power laws** for large incomes, the ancestor of heavy-tail models
- Mandelbrot (1963): cotton prices
  - heavy tails, which the Normal distribution does not describe
- The exponent α of these power laws
  - is the tail index estimated with the Hill method

Back

Cont's Facts in Later Studies

- Ratliff-Crain, Van Oort, Koehler & Tivnan (2025)
  - re-test all 11 facts on intraday returns of the Dow 30 stocks, October 2018 – March 2019
  - conclusive evidence for 8 facts, no support for the remaining 3
- Chakraborti, Toke, Patriarca & Abergel (2011)
  - survey the empirical facts of returns and of the order book, from intraday to lower frequencies

Back

Annual Variance under Autocorrelation (1/2)

- The variance of a sum of h daily returns depends on the autocorrelations

$$\mathrm{Var}\Big(\sum_{j=1}^{h} r_{t+j}\Big) = h\sigma^2\Big[1 + 2\sum_{j=1}^{h-1}(1 - j/h)\rho_j\Big]$$

  - h: the horizon in days; σ^2: the daily variance; ρ_j: the autocorrelation at lag j
  - assumption: weakly stationary returns (constant mean and variance, autocovariances depending only on the lag)
  - without autocorrelation ($\rho_j = 0$) it is $h\sigma^2$, i.e. the $\sqrt{h}$ rule for volatility
- **Variance ratio** $VR(h) = \mathrm{Var}(h\text{-day sum})/(h\sigma^2)$
  - $VR(h) = 1$: no autocorrelation; $VR(h) > 1$: positive autocorrelation; $VR(h) < 1$: negative autocorrelation
  - volatility over horizon h is the $\sqrt{h}$-rule value multiplied by $\sqrt{VR(h)}$

Back

Annual Variance under Autocorrelation (2/2)

- Autoregression of order 1, AR(1), with $\rho_j = \rho^j$: $\sqrt{VR(h)} \to \sqrt{(1+\rho)/(1-\rho)}$ as h grows
  - ρ: the lag-1 autocorrelation, $|\rho| < 1$
  - EUR/RON, $\rho_1 = 0.17$: factor 1.19, annual volatility 4.4% $\to$ 5.3%
  - S&P 500, $\rho_1 = -0.079$: factor 0.92
- Using the sample lag-1 autocorrelation as an AR(1) parameter is an assumption
  - check the whole ACF (Seminar 1, A7)
  - variance-ratio tests: Chapter 2

The Null Hypothesis of JB, and Tests for Time Series

- JB is a Lagrange Multiplier test Jarque & Bera (1980)
  - under i.i.d. returns from the Normal distribution, $\sqrt{T}\,(\hat{S}, \hat{K} - 3) \to N\big(0, \mathrm{diag}(6, 24)\big)$, hence χ^2_2
  - $\mathrm{diag}(6, 24)$: a diagonal covariance matrix; $\hat{S}$ and $\hat{K}$ are asymptotically independent, with variances $6/T$ and $24/T$
  - JB is the sum of the two squared standardised statistics, hence χ^2 with 2 degrees of freedom
- GARCH-type dependence changes $\mathrm{Var}(\hat{S})$ and $\mathrm{Var}(\hat{K})$
  - a rejection mixes the marginal distribution and independence
- Bai & Ng (2005): tests of skewness, kurtosis and normality of the marginal distribution of a time series
  - studentised with HAC variances
  - their separate skewness and kurtosis tests (null $S = 0$ or $K = 3$ without normality) need $\mathbb{E}r^6 < \infty$ and $\mathbb{E}r^8 < \infty$

Back

The Robust Portmanteau Test and Its Assumptions

- $\tilde{Q}(m) = \sum_{k=1}^{m} \hat{\rho}_k^2 / \hat{\tau}_k \sim \chi_m^2$ Lobato, Nankervis & Savin (2001) needs $\mathbb{E}r^4 < \infty$
  - $\sim$: "is distributed as", here asymptotically
  - and zero cross terms: $\mathbb{E}[x_t^2 x_{t-k} x_{t-l}] = 0$ for two different lags $k \neq l$
  - then the $\hat{\rho}_k$ at different lags are asymptotically uncorrelated and the sum is χ_m^2
- Dalla, Giraitis & Phillips (2022) keep only the significant cross terms in the covariance of the $\hat{\rho}_k$
  - none for the S&P 500, BET-TR and Bitcoin
- $\hat{\tau}_k$ is the variance under $\rho_k = 0$
  - to compare a non-zero ρ_1 across periods use the HAC SE of the slope of r_t on r_{t-1} (Seminar 1, B11)
- ACF of $|r_t|$ and of r_t^2: Normal limits need $\alpha > 4$ and $\alpha > 8$ Davis & Mikosch (1998)

Back

Hill Estimator, Asymptotic Distribution and Alternatives

- Hill = maximum likelihood estimator of a Pareto tail above $u = X_{(k+1)}$, conditional on u (Seminar 1, A11)
- If $k \to \infty$ and $k/n \to 0$, and the tail is Pareto up to a second-order term

$$\sqrt{k}\,(\hat{\alpha}_k - \alpha) \;\to\; N(0, \alpha^2)$$

  - second-order term: a correction to the power law that vanishes as $x \to \infty$
  - consequence: $\mathrm{SE}(\hat{\alpha}_k) \approx \alpha/\sqrt{k}$, the formula in the main part
  - if k grows too fast, a bias term appears: the tail is only approximately Pareto
- Dependent data (GARCH-type): $\hat{\alpha}$ stays consistent
  - but extremes cluster and the variance exceeds α^2/k Hsing (1991)
- Choosing k: plot $\hat{\alpha}$ against $\log k$ ("alt-Hill" plot) Drees, Resnick & de Haan (2000)
  - the log scale stretches the region of small k, where the actual tail lies
- Regression alternative Gabaix & Ibragimov (2011)
  - log rank on log size, with rank shifted by $-\frac{1}{2}$: $\ln(i - \frac{1}{2}) = a - b \ln X_{(i)}$, $i = 1, \ldots, k$
  - the slope $\hat{b}$ estimates α; the shift by $\frac{1}{2}$ reduces the small-sample bias
  - $\mathrm{SE} = \hat{\alpha}\sqrt{2/k}$

Back

Two Tails and the Tail of $|r_t|$

- $P(|X| > x) = P(X > x) + P(X < -x)$
  - with indices α_- (losses) and α_+ (gains), $|X|$ has index $\min(\alpha_-, \alpha_+)$
- At the same k the estimates are not comparable
  - the k largest $|r_t|$ come from both tails
  - common threshold u: the lower of the two one-sided thresholds at $k = 2.5\%$ of n

	u (%)	k for $\lvert r_t \rvert$	$\hat{\alpha}$, $\lvert r_t \rvert$	$\hat{\alpha}_-$, losses	$\hat{\alpha}_+$, gains
S&P 500	2.17	508	2.90	2.85	2.96
BET-TR	1.79	153	2.54	2.20	3.06
Bitcoin	7.11	233	3.02	2.71	3.36
EUR/RON	0.54	295	2.28	2.63	2.04
Gold	1.98	513	3.11	2.97	3.28

- At a common u, the $|r_t|$ estimate lies between the two tails
  - Bitcoin's 4.2 at $k = 2.5\%$ reflects second-order bias
  - report both one-sided indices
  - Seminar 1, B6 uses $k = 2\%$ of n per tail: Bitcoin 2.89 / 3.62, BET-TR 2.19 / 3.08

Back

Which Tail Index Is the Chapter's Estimate?

- The chapter uses the fixed reference choice: $|r_t|$ at $k = 2.5\%$ of n (table in the main part)
- The common-threshold k for $|r_t|$ lies outside the 1–5% range of the main part
  - S&P 500 $508/9{,}245 = 5.5\%$, gold $513/9{,}559 = 5.4\%$, EUR/RON $295/5{,}352 = 5.5\%$, Bitcoin 5.3%, BET-TR 5.1%
  - a larger k lets less extreme days in, which can bias $\hat{\alpha}$ (choice of k)
- EUR/RON falls from 2.82 to 2.28
  - more than twice as many tail days (295 versus 133)
  - and the gain tail, $\hat{\alpha}_+ = 2.04$, is the heavier one: $|r_t|$ inherits the heavier tail
  - read the gap as threshold sensitivity, not as a second estimate of the same quantity
- The common-threshold table answers a different question: which tail, losses or gains, is heavier

Estimating d, the Formulas (1/2)

- Fourier frequencies $\lambda_j = 2\pi j/T$, $j = 1, \ldots, m$; the m lowest are used, here $m = T^{0.65}$
- The **periodogram**: the sample estimator of the spectral density

$$I(\lambda_j) = \frac{1}{2\pi T}\left|\sum_{t=1}^{T} x_t\, e^{-\mathrm{i}\lambda_j t}\right|^2$$

  - x_t: the demeaned series (here $|r_t|$ minus its mean); i: the imaginary unit
  - $I(\lambda_j)$: how much of the variation in x_t comes from the cycle of frequency λ_j
- Log-periodogram regression Geweke & Porter-Hudak (1983): regress $\ln I(\lambda_j)$ on $-2\ln(2\sin(\lambda_j/2))$
  - the reason: $\ln f(\lambda) \approx \ln G - 2d \ln \lambda$, and $2\sin(\lambda/2) \approx \lambda$ at low frequencies
  - the slope is $\hat{d}$, SE $= \pi/\sqrt{24m}$

Back

Estimating d, the Formulas (2/2)

- Local Whittle Robinson (1995): $\hat{d}$ minimises an objective built from the approximation $f(\lambda) \approx G\lambda^{-2d}$

$$\hat{d} = \arg\min_{d} \left[\ln\left(\frac{1}{m} \sum_{j=1}^{m} \lambda_j^{2d} I(\lambda_j) \right) - \frac{2d}{m} \sum_{j=1}^{m} \ln \lambda_j \right]$$

  - $\arg\min_d$: the value of d that minimises the expression; the constant G has been eliminated analytically
  - $\sqrt{m}(\hat{d} - d) \to N(0, \frac{1}{4})$, so $\text{SE} = 1/(2\sqrt{m})$
- Local Whittle is robust to non-Gaussianity
  - a larger m lowers the SE
  - but lets short-run (GARCH) dynamics bias $\hat{d}$ (Seminar 1, B12)

Markets in Detail

- **Crypto-assets**: an asset class with its own statistical profile Härdle, Harvey & Reule (2020)
  - their returns have little exposure to the risk factors of traditional markets Liu & Tsyvinski (2021)
  - crypto modelling and the CRIX (CRypto IndeX): Härdle, Chen & Overbeck (2017); Chapter 16
- **EUR/RON**: BNR reference rate from 1 July 2005
  - the new leu (RON, 1 RON = 10,000 old lei)
  - long calm periods interrupted by sudden adjustments
  - EUR/RON market (interbank) quote series showed spurious negative autocorrelation (quote errors)
- **BET-TR**: the bank-asset tax was announced on 18 December 2018
  - adopted on 28 December 2018 as Government Emergency Ordinance (GEO) 114/2018
  - policy shocks produce single-day crashes without comparable rallies
- **Non-synchronous trading**: index constituents do not all trade at every moment
  - so the index reflects news with a delay
  - creating positive autocorrelation Campbell, Lo & MacKinlay (1997), ch. 3

Back

The Maximum Drawdown of a Brownian Motion

- Driftless Brownian motion of the log price with annual volatility σ over T years
  - here T is the horizon in years, not the number of days
- The maximum **log** drawdown $D = -\ln(1 + MDD) \geq 0$, the largest fall in the log price, has

$$\mathbb{E}[D] = \sqrt{\pi/2}\,\sigma\sqrt{T} \qquad \text{Magdon-Ismail, Atiya, Pratap \& Abu-Mostafa (2004)}$$

- $MDD = e^{-D} - 1$; $\sqrt{\pi/2} \approx 1.25$
  - without drift the expected maximum drawdown grows in proportion to $\sqrt{T}$
  - with a positive drift μ, $\mathbb{E}[D] \approx \frac{2\sigma^2}{\mu}\left[\frac{1}{4}\ln\frac{\mu^2 T}{2\sigma^2} + 0.49\right]$ grows only like $\ln T$
- Gold 1990–2026: $\sigma = 16.3\%$, $T = 36.7$
  - without drift $\mathbb{E}[D] = 1.24$, twice the observed 0.592 (-44.7%) (Seminar 1, A2(b–e))
  - with the observed drift, $\mu = 6.5\%$ a year: $\mathbb{E}[D] \approx 0.62$ (formula) and 0.64 (simulation), close to the observed value
- Drawdowns and Calmar ratios from samples of different length are not comparable

Back

Sharpe Ratios beyond the Normal Distribution

- i.i.d., any distribution with $\mathbb{E}r^4 < \infty$ Opdyke (2007)
  - $\text{Var}(\widehat{SR}_d) \approx (1 - S\,SR_d + \frac{K-1}{4} SR_d^2)/T$
  - S, K: the skewness and kurtosis of daily returns; for the Normal distribution ($S = 0$, $K = 3$) this gives Lo's formula
  - negative skewness and high kurtosis raise the variance of $\widehat{SR}_d$ when $SR_d > 0$
  - here the SEs change by at most 0.02
- The Deflated Sharpe Ratio corrects for non-normality and multiple testing (Chapter 13)

	CAGR	Sharpe	Sortino	Calmar
S&P 500	8.7%	0.55	0.78	0.15
BET-TR	21.2%	1.34	1.87	0.68
Bitcoin	53.9%	0.98	1.45	0.65
EUR/RON	1.8%	0.43	0.65	0.12
Gold	6.7%	0.48	0.69	0.15

MFM_ch1_risk_measures

Back

Case Study, Estimation Details

- Targets (Sections 3.4–3.5 of the paper)
  - the 30-day implied volatility index of each index and the 9-day VIX
  - realised volatility of day $t+1$
  - volatility in decimals (0.03 = 3 volatility points)
- Kernels, eq. (3.9) in the paper: $K(\tau) = Z_{\alpha,\delta}^{-\mathbf{1}}(\tau+\delta)^{-\alpha}$
  - normalised so that $\sum_i K(i\Delta t)\Delta t = 1$ on the 1,000-day window
  - i: the age of the day, in days; $\Delta t = 1/252$: one day, in years
- Estimation: $(\beta_{\mathbf{0}}, \beta_{\mathbf{1}}, \beta_{\mathbf{2}}, \alpha_{\mathbf{1}}, \delta_{\mathbf{1}}, \alpha_{\mathbf{2}}, \delta_{\mathbf{2}})$ by nonlinear least squares on the training set
  - initialised using a lasso regression on features built with exponential kernels
- Our replication, started from the values in Table 3 of the paper (in parentheses)
  - $\hat{\beta}_{\mathbf{0}} = 0.052$ (0.057), $\hat{\beta}_{\mathbf{1}} = -0.090$ (-0.095)
  - $\hat{\beta}_{\mathbf{2}} = 0.868$ (0.82), $\hat{\alpha}_{\mathbf{1}} = 1.13$ (1.06)
- The 9-day VIX exists only from 2011 (shorter training set)
- The restricted models ($\beta_{\mathbf{1}} = 0$, $\beta_{\mathbf{2}} = 0$) are Seminar 1, C3

Back

Case Study, R^2 for All Indices (1/2)

Index	Implied volatility (paper, Table 2) R^2 train	R^2 test	Realised volatility (paper, Table 4) R^2 train	R^2 test
S&P 500 (VIX)	0.946	0.855	0.738	0.654
S&P 500 (9-day VIX)	0.876	0.914	–	–
Euro Stoxx 50	0.929	0.913	0.672	0.682
FTSE 100	0.925	0.870	0.650	0.617
DAX	0.934	0.918	0.722	0.557
Nikkei 225	0.890	0.800	0.563	0.504

- VIX parameters: $\beta_0 = 0.057$, $\beta_1 = -0.095$, $\beta_2 = 0.82$
- Across all targets $\alpha_2 \in [1.2, 2.3]$ (Tables 3 and 5 of the paper)
  - long memory near criticality (Section 3.2 of the paper)

Back

Case Study, R^2 for All Indices (2/2)

- Dropping Σ_t hurts the most
- The model beats two ARCH benchmarks and four published path-dependent models
  - Figures 3.8–3.9 and Appendix C of the paper
- Beyond this course (optional): its Markovian 4-factor version, driven by Brownian motion (Section 4.6 and Table 9 of the paper)
  - Markovian: the next value depends only on the current state (here, 4 factors), not on the whole path
  - gives estimated values of about 0.12–0.17 of the Hurst exponent H for daily realised volatility, as in data
  - H: the roughness exponent of a path; Brownian motion has $H = 0.5$, and $H < 0.5$ means a path "rougher" than Brownian motion
  - although it has no rough volatility ($H < 0.5$ built into the model), the Markovian version reproduces the small values of H estimated from daily data

The Discovery Loop, Reusable Prompts

Literature

- Find peer-reviewed work since 2015 on time variation or structural breaks in tail indices of asset returns, with tests valid under GARCH-type dependence.
- For each: DOI, estimator, test, data.
- Flag entries you are unsure of.

Hypothesis

- State competing hypotheses for a change in Bitcoin's tail after the 2024 spot ETFs.
- What does each imply for the Hill estimate of the tail index of raw and of GARCH-standardised returns?

Data and code

- Use daily Bitcoin closing prices in USD; log returns on its own calendar;
- Hill estimate of the tail index for $|r|$, with k = 2.5% of n, before and after 11 January 2024; moving-block bootstrap standard errors.

Robustness

- Write a pre-registered plan: k from 1% to 5% of n, loss and gain tails, GARCH-standardised residuals, block lengths 10 to 50 days, a placebo date and a placebo asset.

Critique

- Referee this design.
- With 24 tail observations after the break, what is the power against a change of 1 in the tail index?

Back

Mandatory Checks for the Researcher

- **Wrong return or price series**: reproduce the chapter's Hill table before anything else
- **Choice of k after seeing the result**
  - fix the rule ($k = 2.5\%$ of n) in advance
  - show the Hill plot for 1–5%
- **i.i.d. standard errors**: $\hat{\alpha}/\sqrt{k}$ ignores volatility clustering
  - compare with a block bootstrap
  - for S&P 500, 17 Sep 2014–10 Jan 2024: bootstrap SE 0.817 versus i.i.d. 0.370, hence 2.2 times larger
- **Break date chosen from the data**
  - use a date fixed by the event
  - or a sup-type change-point test with its own critical values Hoga (2017)
- **Bad ticks create fake tails**
  - list the ten largest $|r_t|$ and match each to a dated event
- **Low power**: report the minimum detectable change
  - 2.8 SEs of the difference for a two-sided 5% test with 80% power
  - 2.5 SEs for a one-sided test

Back

Practical Checklist for Empirical Work

- **Know your data**
  - price vs total-return index
  - adjusted vs raw prices
  - fixing vs traded price
  - official reference rates where possible (EUR/RON)
- **Clean before you model**
  - non-trading days, holidays, stale quotes, obvious errors
  - match the largest moves to real events
- **Use log returns** for single-asset dynamics, simple returns for portfolios
- **Report robust statistics**: quantiles, the number of tail days and the tail index $\hat{\alpha}$ next to kurtosis
- **Annualise carefully**
  - 252 vs 365 periods
  - the $\sqrt{A}$ rule needs uncorrelated returns with constant variance
- **Reproducibility**: code and data on GitHub, as Quantlets

Back

Further Study on Quantinar

Advanced courses on quantinar.com

- Statistics of Financial Markets
- Tukey's g- and h- transformations
- Applied Time Series Analysis with Python

Code, data and recommended reading

- All charts and tables: github.com/danpele/MFM/tree/main/Quantlets/Ch_01
- Recommended reading: Cont (2001); Franke, Härdle & Hafner (2019); Campbell, Lo & MacKinlay (1997), ch. 1–2

Back

Acronyms used in this material (1/2)

- **ACF**: AutoCorrelation Function
- **AI**: Artificial Intelligence
- **AR**: AutoRegressive (model)
- **ARCH**: AutoRegressive Conditional Heteroskedasticity
- **ARCH-LM**: ARCH Lagrange Multiplier (test)
- **BET-TR**: BET Total Return (index)
- **BNR**: Banca Națională a României — *the National Bank of Romania*
- **BVB**: Bursa de Valori București — *the Bucharest Stock Exchange*
- **CAGR**: Compound Annual Growth Rate
- **CI**: Confidence Interval
- **CLT**: Central Limit Theorem
- **CRIX**: CRypto IndeX
- **DAX**: Deutscher Aktienindex — *the German stock index*
- **EGARCH**: Exponential GARCH
- **ETF**: Exchange-Traded Fund
- **FTSE**: Financial Times Stock Exchange (FTSE Russell, index provider)
- **GARCH**: Generalised AutoRegressive Conditional Heteroskedasticity

Acronyms used in this material (2/2)

- **GEO**: Government Emergency Ordinance
- **GJR-GARCH**: Glosten–Jagannathan–Runkle GARCH
- **HAC**: Heteroskedasticity and Autocorrelation Consistent (standard errors)
- **HAR-RV**: Heterogeneous AutoRegressive model of Realised Volatility
- **JB**: Jarque–Bera (test)
- **LB**: Ljung–Box (test)
- **LM**: Lagrange Multiplier
- **MDD**: Maximum DrawDown
- **OHLC**: Open, High, Low, Close
- **PDV**: Path-Dependent Volatility
- **QQ**: Quantile–Quantile (plot)
- **RMSE**: Root Mean Squared Error
- **SE**: Standard Error
- **TSPL**: Time-Shifted Power Law (kernel)
- **US**: United States
- **VIX**: Cboe Volatility IndeX
- **VIX9D**: Cboe 9-Day Volatility Index

References (1/4)

- Andersen, T.G., Bollerslev, T., Diebold, F.X., Ebens, H. (2001). The distribution of realized stock return volatility. *Journal of Financial Economics*, 61(1), 43–76.
- Bachelier, L. (1900). Théorie de la spéculation. *Annales scientifiques de l'École Normale Supérieure*, 17, 21–86.
- Bai, J., Ng, S. (2005). Tests for skewness, kurtosis, and normality for time series data. *Journal of Business & Economic Statistics*, 23(1), 49–60.
- Bollerslev, T., Hood, B., Huss, J., Pedersen, L.H. (2018). Risk everywhere: modeling and managing volatility. *Review of Financial Studies*, 31(7), 2729–2773.
- Bouchaud, J.-P., Matacz, A., Potters, M. (2001). Leverage effect in financial markets: the retarded volatility model. *Physical Review Letters*, 87(22), 228701.
- Campbell, J.Y., Lo, A.W., MacKinlay, A.C. (1997). The Econometrics of Financial Markets. Princeton University Press.
- Chakraborti, A., Toke, I.M., Patriarca, M., Abergel, F. (2011). Econophysics review: I. Empirical facts. *Quantitative Finance*, 11(7), 991–1012.
- Christie, A.A. (1982). The stochastic behavior of common stock variances: value, leverage and interest rate effects. *Journal of Financial Economics*, 10(4), 407–432.
- Clark, P.K. (1973). A subordinated stochastic process model with finite variance for speculative prices. *Econometrica*, 41(1), 135–155.
- Cong, L.W., Li, X., Tang, K., Yang, Y. (2023). Crypto wash trading. *Management Science*, 69(11), 6427–6454.
- Cont, R. (2001). Empirical properties of asset returns: stylized facts and statistical issues. *Quantitative Finance*, 1(2), 223–236.
- Dalla, V., Giraitis, L., Phillips, P.C.B. (2022). Robust tests for white noise and cross-correlation. *Econometric Theory*, 38(5), 913–941.
- Davis, R.A., Mikosch, T. (1998). The sample autocorrelations of heavy-tailed processes with applications to ARCH. *Annals of Statistics*, 26(5), 2049–2080.
- Diebold, F.X., Inoue, A. (2001). Long memory and regime switching. *Journal of Econometrics*, 105(1), 131–159.
- Ding, Z., Granger, C.W.J., Engle, R.F. (1993). A long memory property of stock market returns and a new model. *Journal of Empirical Finance*, 1(1), 83–106.

References (2/4)

- Drees, H., Resnick, S., de Haan, L. (2000). How to make a Hill plot. *Annals of Statistics*, 28(1), 254–274.
- Engle, R.F. (1982). Autoregressive conditional heteroscedasticity with estimates of the variance of United Kingdom inflation. *Econometrica*, 50(4), 987–1007.
- Fama, E.F. (1965). The behavior of stock-market prices. *Journal of Business*, 38(1), 34–105.
- Franke, J., Härdle, W.K., Hafner, C.M. (2019). Statistics of Financial Markets: An Introduction, 5th ed. Springer.
- Gabaix, X., Ibragimov, R. (2011). Rank − 1/2: a simple way to improve the OLS estimation of tail exponents. *Journal of Business & Economic Statistics*, 29(1), 24–39.
- Garman, M.B., Klass, M.J. (1980). On the estimation of security price volatilities from historical data. *Journal of Business*, 53(1), 67–78.
- Geweke, J., Porter-Hudak, S. (1983). The estimation and application of long memory time series models. *Journal of Time Series Analysis*, 4(4), 221–238.
- Granger, C.W.J., Ding, Z. (1995). Some properties of absolute return: an alternative measure of risk. *Annales d'Économie et de Statistique*, 40, 67–91.
- Guyon, J., Lekeufack, J. (2023). Volatility is (mostly) path-dependent. *Quantitative Finance*, 23(9), 1221–1258.
- Härdle, W.K., Chen, C.Y.-H., Overbeck, L. (Eds.) (2017). Applied Quantitative Finance, 3rd ed. Springer.
- Härdle, W.K., Harvey, C.R., Reule, R.C.G. (2020). Understanding cryptocurrencies. *Journal of Financial Econometrics*, 18(2), 181–208.
- Hill, B.M. (1975). A simple general approach to inference about the tail of a distribution. *Annals of Statistics*, 3(5), 1163–1174.
- Hoga, Y. (2017). Change point tests for the tail index of β-mixing random variables. *Econometric Theory*, 33(4), 915–954.
- Hsing, T. (1991). On tail index estimation using dependent data. *Annals of Statistics*, 19(3), 1547–1569.

References (3/4)

- Inclán, C., Tiao, G.C. (1994). Use of cumulative sums of squares for retrospective detection of changes of variance. *Journal of the American Statistical Association*, 89(427), 913–923.
- Jarque, C.M., Bera, A.K. (1980). Efficient tests for normality, homoscedasticity and serial independence of regression residuals. *Economics Letters*, 6(3), 255–259.
- Karpoff, J.M. (1987). The relation between price changes and trading volume: a survey. *Journal of Financial and Quantitative Analysis*, 22(1), 109–126.
- Kim, T.-H., White, H. (2004). On more robust estimation of skewness and kurtosis. *Finance Research Letters*, 1(1), 56–73.
- Ledoit, O., Wolf, M. (2008). Robust performance hypothesis testing with the Sharpe ratio. *Journal of Empirical Finance*, 15(5), 850–859.
- Liu, Y., Tsyvinski, A. (2021). Risks and returns of cryptocurrency. *Review of Financial Studies*, 34(6), 2689–2727.
- Ljung, G.M., Box, G.E.P. (1978). On a measure of lack of fit in time series models. *Biometrika*, 65(2), 297–303.
- Lo, A.W. (2002). The statistics of Sharpe ratios. *Financial Analysts Journal*, 58(4), 36–52.
- Lobato, I., Nankervis, J.C., Savin, N.E. (2001). Testing for autocorrelation using a modified Box–Pierce Q test. *International Economic Review*, 42(1), 187–205.
- Magdon-Ismail, M., Atiya, A.F., Pratap, A., Abu-Mostafa, Y.S. (2004). On the maximum drawdown of a Brownian motion. *Journal of Applied Probability*, 41(1), 147–161.
- Mandelbrot, B. (1963). The variation of certain speculative prices. *Journal of Business*, 36(4), 394–419.
- Mikosch, T., Stărică, C. (2004). Nonstationarities in financial time series, the long-range dependence, and the IGARCH effects. *Review of Economics and Statistics*, 86(1), 378–390.
- Opdyke, J.D. (2007). Comparing Sharpe ratios: so where are the p-values?. *Journal of Asset Management*, 8(5), 308–336.
- Parkinson, M. (1980). The extreme value method for estimating the variance of the rate of return. *Journal of Business*, 53(1), 61–65.

References (4/4)

- Praetz, P.D. (1972). The distribution of share price changes. *Journal of Business*, 45(1), 49–55.
- Ratliff-Crain, E., Van Oort, C.M., Koehler, M., Tivnan, B.F. (2025). Revisiting Cont's stylized facts for modern stock markets. *Quantitative Finance*, 25(9), 1343–1373.
- Robinson, P.M. (1995). Gaussian semiparametric estimation of long range dependence. *Annals of Statistics*, 23(5), 1630–1661.
- Rogers, L.C.G., Satchell, S.E. (1991). Estimating variance from high, low and closing prices. *Annals of Applied Probability*, 1(4), 504–512.
- Romano, J.P., Thombs, L.A. (1996). Inference for autocorrelations under weak assumptions. *Journal of the American Statistical Association*, 91(434), 590–600.
- Sharpe, W.F. (1966). Mutual fund performance. *Journal of Business*, 39(1), 119–138.
- Sharpe, W.F. (1994). The Sharpe ratio. *Journal of Portfolio Management*, 21(1), 49–58.
- Sortino, F.A., van der Meer, R. (1991). Downside risk. *Journal of Portfolio Management*, 17(4), 27–31.
- Student (1908). The probable error of a mean. *Biometrika*, 6(1), 1–25.
- Taylor, S.J. (1986). Modelling Financial Time Series. Wiley; 2nd ed. (2008), World Scientific, doi:10.1142/6578.
- Yang, D., Zhang, Q. (2000). Drift-independent volatility estimation based on high, low, open, and close prices. *Journal of Business*, 73(3), 477–492.